

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.5	4	0.7	0	0	81.8
U2 - Beta(0.5, 0.5)	13.4	4.2	0.6	0	0	81.8
U2 - Beta(0.7, 0.3)	12.9	4.6	0.6	0	0	81.9
U2 - Beta(1, 1)	12.9	4.2	0.8	0	0	82.1
U3 - Beta(0.3, 0.7)	13.9	3.6	0.6	0	0	81.9
U3 - Beta(0.5, 0.5)	13.9	3.3	0.7	0	0	82.1
U3 - Beta(0.7, 0.3)	14.1	3.2	0.5	0	0	82.2
U3 - Beta(1, 1)	14	3.6	0.4	0	0	82
Mean administrations						
U2 - Beta(0.3, 0.7)	11.5	2.9	1.3	0.4	0.1	-
U2 - Beta(0.5, 0.5)	11.5	3	1.3	0.3	0.1	-
U2 - Beta(0.7, 0.3)	11.5	3	1.2	0.3	0.1	-
U2 - Beta(1, 1)	11.5	3	1.2	0.3	0.1	-
U3 - Beta(0.3, 0.7)	11.5	2.9	1.3	0.3	0.1	-
U3 - Beta(0.5, 0.5)	11.5	2.9	1.3	0.3	0.1	-
U3 - Beta(0.7, 0.3)	11.5	3	1.3	0.3	0.1	-
U3 - Beta(1, 1)	11.5	3	1.2	0.3	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.4	1.6	0.5	0.2	0	-
U2 - Beta(0.5, 0.5)	3.4	1.6	0.5	0.2	0	-
U2 - Beta(0.7, 0.3)	3.4	1.6	0.4	0.2	0	-
U2 - Beta(1, 1)	3.4	1.6	0.4	0.2	0	-
U3 - Beta(0.3, 0.7)	3.4	1.6	0.5	0.2	0	-
U3 - Beta(0.5, 0.5)	3.4	1.6	0.5	0.2	0	-
U3 - Beta(0.7, 0.3)	3.4	1.6	0.5	0.2	0	-
U3 - Beta(1, 1)	3.4	1.6	0.4	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						574.9
U2 - Beta(0.5, 0.5)						574.8
U2 - Beta(0.7, 0.3)						574.5
U2 - Beta(1, 1)						574.7
U3 - Beta(0.3, 0.7)						573.7
U3 - Beta(0.5, 0.5)						573.7
U3 - Beta(0.7, 0.3)						574.2
U3 - Beta(1, 1)						574.7
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	37.8	19.5	30.2	7.3	1.1	4.1
U2 - Beta(0.5, 0.5)	37.8	19.8	30.7	6.4	1.2	4.1
U2 - Beta(0.7, 0.3)	37.8	20.3	30.1	6.9	0.8	4.1
U2 - Beta(1, 1)	37.5	20.4	29.7	7.4	0.9	4.1
U3 - Beta(0.3, 0.7)	40	19.7	28.8	6.6	0.8	4.1
U3 - Beta(0.5, 0.5)	40.4	21.3	27.7	6.1	0.5	4
U3 - Beta(0.7, 0.3)	40.7	21.5	26.6	6.5	0.6	4.1
U3 - Beta(1, 1)	41.6	20.8	26.5	6.4	0.7	4
Mean administrations						
U2 - Beta(0.3, 0.7)	12.5	5.6	6.9	3.2	1.2	-
U2 - Beta(0.5, 0.5)	12.5	5.7	6.8	3.2	1.2	-
U2 - Beta(0.7, 0.3)	12.5	5.7	6.8	3.1	1.1	-
U2 - Beta(1, 1)	12.5	5.7	6.8	3.2	1.2	-
U3 - Beta(0.3, 0.7)	12.5	5.5	7	3.1	1.2	-
U3 - Beta(0.5, 0.5)	12.5	5.7	6.8	3.2	1.2	-
U3 - Beta(0.7, 0.3)	12.5	5.7	6.8	3.1	1.1	-
U3 - Beta(1, 1)	12.5	5.7	6.7	3.2	1.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.7	2.6	1.9	0.9	0.3	-
U2 - Beta(0.5, 0.5)	3.7	2.6	1.9	0.9	0.3	-
U2 - Beta(0.7, 0.3)	3.7	2.6	1.9	0.8	0.3	-
U2 - Beta(1, 1)	3.7	2.6	1.9	0.9	0.3	-
U3 - Beta(0.3, 0.7)	3.7	2.6	1.9	0.8	0.3	-
U3 - Beta(0.5, 0.5)	3.7	2.6	1.9	0.9	0.3	-
U3 - Beta(0.7, 0.3)	3.7	2.6	1.9	0.8	0.3	-
U3 - Beta(1, 1)	3.7	2.6	1.9	0.9	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1095.3
U2 - Beta(0.5, 0.5)						1094.8
U2 - Beta(0.7, 0.3)						1094
U2 - Beta(1, 1)						1095.1
U3 - Beta(0.3, 0.7)						1095.5
U3 - Beta(0.5, 0.5)						1096
U3 - Beta(0.7, 0.3)						1094.2
U3 - Beta(1, 1)						1095

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	15.1	41.7	18.5	5.8	0.5	18.4
U2 - Beta(0.5, 0.5)	15.4	41.3	18.1	6.2	0.6	18.4
U2 - Beta(0.7, 0.3)	15.3	41.7	17.8	5.7	0.9	18.6
U2 - Beta(1, 1)	15.5	41.4	18	5.8	0.8	18.5
U3 - Beta(0.3, 0.7)	16.6	42.6	16.2	5.5	0.6	18.5
U3 - Beta(0.5, 0.5)	16.9	44.4	14.7	4.9	0.5	18.6
U3 - Beta(0.7, 0.3)	17.1	44.8	14.9	4.3	0.3	18.6
U3 - Beta(1, 1)	16.5	45.5	14.6	4.5	0.5	18.4
Mean administrations						
U2 - Beta(0.3, 0.7)	9.5	7.8	5.8	2.9	1.1	-
U2 - Beta(0.5, 0.5)	9.4	7.9	5.7	2.9	1.2	-
U2 - Beta(0.7, 0.3)	9.5	8	5.6	2.8	1.1	-
U2 - Beta(1, 1)	9.5	7.8	5.7	2.9	1.2	-
U3 - Beta(0.3, 0.7)	9.4	7.8	5.8	3	1.1	-
U3 - Beta(0.5, 0.5)	9.5	7.8	5.8	2.9	1.1	-
U3 - Beta(0.7, 0.3)	9.5	8	5.6	2.8	1.1	-
U3 - Beta(1, 1)	9.4	7.9	5.7	2.9	1.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.7	3.4	1.6	0.9	0.3	-
U2 - Beta(0.5, 0.5)	2.7	3.4	1.6	0.9	0.3	-
U2 - Beta(0.7, 0.3)	2.7	3.4	1.6	0.9	0.3	-
U2 - Beta(1, 1)	2.7	3.4	1.6	0.9	0.4	-
U3 - Beta(0.3, 0.7)	2.6	3.3	1.6	0.9	0.3	-
U3 - Beta(0.5, 0.5)	2.7	3.3	1.6	0.9	0.3	-
U3 - Beta(0.7, 0.3)	2.7	3.4	1.6	0.9	0.3	-
U3 - Beta(1, 1)	2.7	3.4	1.6	0.9	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						998
U2 - Beta(0.5, 0.5)						997.4
U2 - Beta(0.7, 0.3)						997.3
U2 - Beta(1, 1)						997.1
U3 - Beta(0.3, 0.7)						997.4
U3 - Beta(0.5, 0.5)						997.1
U3 - Beta(0.7, 0.3)						997.7
U3 - Beta(1, 1)						997.4
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.4	10	26	41.1	11.4	4.1
U2 - Beta(0.5, 0.5)	7.8	10.5	24.7	42	10.9	4.1
U2 - Beta(0.7, 0.3)	9.3	9.8	23.2	43.2	10.6	3.9
U2 - Beta(1, 1)	9.8	10.2	22.5	42.5	10.9	4.1
U3 - Beta(0.3, 0.7)	9.9	11	24.1	40.8	10.3	3.9
U3 - Beta(0.5, 0.5)	12.1	11.5	22.3	40.7	9.5	3.9
U3 - Beta(0.7, 0.3)	14	11.5	21.7	39.7	9.1	4
U3 - Beta(1, 1)	14.4	12	21.2	40.1	8.4	3.9
Mean administrations						
U2 - Beta(0.3, 0.7)	4.5	4.4	7.2	8.2	5	-
U2 - Beta(0.5, 0.5)	4.5	4.5	7.2	8.2	4.9	-
U2 - Beta(0.7, 0.3)	4.5	4.6	7.1	8.2	4.9	-
U2 - Beta(1, 1)	4.5	4.6	7.1	8	5.1	-
U3 - Beta(0.3, 0.7)	4.5	4.5	7.1	8.1	5.1	-
U3 - Beta(0.5, 0.5)	4.5	4.5	7.3	8.1	4.9	-
U3 - Beta(0.7, 0.3)	4.6	4.6	7.1	8.1	4.9	-
U3 - Beta(1, 1)	4.5	4.6	7.1	8.1	5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	2.8	2.3	2.7	1.2	-
U2 - Beta(0.5, 0.5)	0.7	2.8	2.3	2.7	1.2	-
U2 - Beta(0.7, 0.3)	0.7	2.9	2.2	2.7	1.2	-
U2 - Beta(1, 1)	0.7	2.9	2.2	2.6	1.2	-
U3 - Beta(0.3, 0.7)	0.7	2.9	2.2	2.7	1.2	-
U3 - Beta(0.5, 0.5)	0.7	2.9	2.3	2.7	1.2	-
U3 - Beta(0.7, 0.3)	0.7	2.9	2.3	2.7	1.2	-
U3 - Beta(1, 1)	0.7	2.9	2.2	2.7	1.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1077.1
U2 - Beta(0.5, 0.5)						1076.2
U2 - Beta(0.7, 0.3)						1076.1
U2 - Beta(1, 1)						1077.8
U3 - Beta(0.3, 0.7)						1078.8
U3 - Beta(0.5, 0.5)						1077.1
U3 - Beta(0.7, 0.3)						1075.7
U3 - Beta(1, 1)						1076.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.6	33.6	27.5	19.5	8.7	7.1
U2 - Beta(0.5, 0.5)	4	35.3	26.7	20	6.9	7.1
U2 - Beta(0.7, 0.3)	5.3	35.7	25.6	19.2	7.1	7.1
U2 - Beta(1, 1)	5	37.3	26.9	17.4	6.3	7.1
U3 - Beta(0.3, 0.7)	5.6	37.9	26.9	18.1	4.4	7.1
U3 - Beta(0.5, 0.5)	6.7	40.8	24.8	16.3	4.3	7.1
U3 - Beta(0.7, 0.3)	7.4	43.1	24.2	14.3	3.9	7.1
U3 - Beta(1, 1)	8.3	43.7	22.6	15.3	3	7.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.6	6.5	6.9	6	4.7	-
U2 - Beta(0.5, 0.5)	4.6	6.7	6.6	6.1	4.6	-
U2 - Beta(0.7, 0.3)	4.7	6.8	6.7	5.9	4.5	-
U2 - Beta(1, 1)	4.7	6.8	6.7	5.9	4.6	-
U3 - Beta(0.3, 0.7)	4.6	6.5	6.8	6	4.7	-
U3 - Beta(0.5, 0.5)	4.6	6.8	6.7	5.9	4.6	-
U3 - Beta(0.7, 0.3)	4.7	6.8	6.8	5.8	4.5	-
U3 - Beta(1, 1)	4.7	6.8	6.7	6	4.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.5	2.2	2.3	1.7	-
U2 - Beta(0.5, 0.5)	0.7	3.5	2.1	2.3	1.6	-
U2 - Beta(0.7, 0.3)	0.7	3.6	2.1	2.2	1.6	-
U2 - Beta(1, 1)	0.8	3.6	2.1	2.2	1.6	-
U3 - Beta(0.3, 0.7)	0.7	3.5	2.1	2.3	1.7	-
U3 - Beta(0.5, 0.5)	0.7	3.6	2.1	2.2	1.6	-
U3 - Beta(0.7, 0.3)	0.7	3.6	2.1	2.2	1.6	-
U3 - Beta(1, 1)	0.7	3.6	2.1	2.3	1.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1007.6
U2 - Beta(0.5, 0.5)						1006.6
U2 - Beta(0.7, 0.3)						1005.8
U2 - Beta(1, 1)						1006.3
U3 - Beta(0.3, 0.7)						1007.4
U3 - Beta(0.5, 0.5)						1006.5
U3 - Beta(0.7, 0.3)						1005.4
U3 - Beta(1, 1)						1006
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	66.9	8.5	2	0.7	0.1	21.8
U2 - Beta(0.5, 0.5)	66.1	9.5	2.2	0.3	0.1	21.8
U2 - Beta(0.7, 0.3)	66.5	9.5	1.7	0.4	0.1	21.8
U2 - Beta(1, 1)	66.8	9	1.5	0.7	0.1	21.9
U3 - Beta(0.3, 0.7)	67.7	8.5	1.6	0.4	0.1	21.7
U3 - Beta(0.5, 0.5)	67.5	9	1.6	0.2	0	21.7
U3 - Beta(0.7, 0.3)	67.7	8.6	1.7	0.1	0.1	21.8
U3 - Beta(1, 1)	68.2	8.9	1	0.3	0.1	21.5
Mean administrations						
U2 - Beta(0.3, 0.7)	20.1	3.6	1.7	0.6	0.2	-
U2 - Beta(0.5, 0.5)	20.1	3.7	1.6	0.6	0.2	-
U2 - Beta(0.7, 0.3)	20.2	3.6	1.6	0.5	0.2	-
U2 - Beta(1, 1)	20.1	3.6	1.7	0.5	0.2	-
U3 - Beta(0.3, 0.7)	20.1	3.6	1.7	0.6	0.2	-
U3 - Beta(0.5, 0.5)	20.1	3.6	1.7	0.6	0.2	-
U3 - Beta(0.7, 0.3)	20.1	3.7	1.6	0.5	0.2	-
U3 - Beta(1, 1)	20.2	3.6	1.6	0.6	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.6	1.4	0.5	0.2	0	-
U2 - Beta(0.5, 0.5)	5.6	1.5	0.5	0.2	0	-
U2 - Beta(0.7, 0.3)	5.6	1.5	0.5	0.2	0	-
U2 - Beta(1, 1)	5.6	1.5	0.5	0.2	0	-
U3 - Beta(0.3, 0.7)	5.6	1.4	0.5	0.2	0	-
U3 - Beta(0.5, 0.5)	5.6	1.5	0.5	0.2	0	-
U3 - Beta(0.7, 0.3)	5.6	1.5	0.5	0.2	0	-
U3 - Beta(1, 1)	5.6	1.5	0.5	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1010.6
U2 - Beta(0.5, 0.5)						1009.8
U2 - Beta(0.7, 0.3)						1012
U2 - Beta(1, 1)						1011.8
U3 - Beta(0.3, 0.7)						1011.4
U3 - Beta(0.5, 0.5)						1010.8
U3 - Beta(0.7, 0.3)						1011.7
U3 - Beta(1, 1)						1011.9

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	29.5	39.5	17.1	7.4	1.8	4.7
U2 - Beta(0.5, 0.5)	29.4	39.1	18.1	7.1	1.4	4.9
U2 - Beta(0.7, 0.3)	28.6	41.1	17.2	6.8	1.4	4.9
U2 - Beta(1, 1)	28.6	40.5	17.1	7.4	1.5	4.9
U3 - Beta(0.3, 0.7)	30.2	41.6	15.5	6.6	1.3	4.8
U3 - Beta(0.5, 0.5)	29.2	44.1	14.7	5.9	1.2	4.9
U3 - Beta(0.7, 0.3)	30.5	42.8	15	5.8	1	4.9
U3 - Beta(1, 1)	30.1	43.3	14.6	6	1	5
Mean administrations						
U2 - Beta(0.3, 0.7)	11.8	7.4	5.3	3.1	1.6	-
U2 - Beta(0.5, 0.5)	11.8	7.5	5.2	3.1	1.5	-
U2 - Beta(0.7, 0.3)	11.8	7.4	5.3	3	1.5	-
U2 - Beta(1, 1)	11.7	7.6	5.3	3	1.6	-
U3 - Beta(0.3, 0.7)	11.8	7.4	5.2	3	1.6	-
U3 - Beta(0.5, 0.5)	11.8	7.5	5.3	3	1.6	-
U3 - Beta(0.7, 0.3)	11.8	7.5	5.3	3	1.5	-
U3 - Beta(1, 1)	11.7	7.4	5.3	3	1.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.4	3	1.3	0.9	0.5	-
U2 - Beta(0.5, 0.5)	3.4	3	1.3	0.9	0.4	-
U2 - Beta(0.7, 0.3)	3.4	3	1.3	0.9	0.4	-
U2 - Beta(1, 1)	3.3	3	1.3	0.9	0.4	-
U3 - Beta(0.3, 0.7)	3.4	3	1.3	0.9	0.5	-
U3 - Beta(0.5, 0.5)	3.4	3	1.3	0.9	0.4	-
U3 - Beta(0.7, 0.3)	3.4	3	1.3	0.9	0.4	-
U3 - Beta(1, 1)	3.4	3	1.3	0.9	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1110.3
U2 - Beta(0.5, 0.5)						1108.9
U2 - Beta(0.7, 0.3)						1108.5
U2 - Beta(1, 1)						1109.2
U3 - Beta(0.3, 0.7)						1109
U3 - Beta(0.5, 0.5)						1109.3
U3 - Beta(0.7, 0.3)						1108.5
U3 - Beta(1, 1)						1108.6

Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	27.6	5.1	1.1	0.1	0	66.1
U2 - Beta(0.5, 0.5)	27.8	4.9	1.1	0.1	0	66.1
U2 - Beta(0.7, 0.3)	28.1	5.2	0.7	0.1	0	65.9
U2 - Beta(1, 1)	27.9	5.5	0.4	0.1	0	66.1
U3 - Beta(0.3, 0.7)	27.7	5.1	0.9	0.1	0	66.2
U3 - Beta(0.5, 0.5)	28.6	4.9	0.5	0.1	0	65.9
U3 - Beta(0.7, 0.3)	28.4	4.9	0.3	0.1	0	66.3
U3 - Beta(1, 1)	29	4.5	0.3	0.1	0	66.1
Mean administrations						
U2 - Beta(0.3, 0.7)	14.6	2.7	1.1	0.2	0.1	-
U2 - Beta(0.5, 0.5)	14.6	2.7	1.1	0.2	0.1	-
U2 - Beta(0.7, 0.3)	14.7	2.8	1	0.2	0.1	-
U2 - Beta(1, 1)	14.7	2.8	1	0.2	0.1	-
U3 - Beta(0.3, 0.7)	14.6	2.8	1.1	0.3	0.1	-
U3 - Beta(0.5, 0.5)	14.7	2.8	1.1	0.2	0.1	-
U3 - Beta(0.7, 0.3)	14.7	2.8	1	0.2	0.1	-
U3 - Beta(1, 1)	14.7	2.7	1.1	0.2	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.4	1.4	0.4	0.1	0	-
U2 - Beta(0.5, 0.5)	4.4	1.4	0.4	0.1	0	-
U2 - Beta(0.7, 0.3)	4.4	1.4	0.4	0.1	0	-
U2 - Beta(1, 1)	4.4	1.4	0.4	0.1	0	-
U3 - Beta(0.3, 0.7)	4.4	1.4	0.4	0.1	0	-
U3 - Beta(0.5, 0.5)	4.4	1.4	0.4	0.1	0	-
U3 - Beta(0.7, 0.3)	4.4	1.4	0.4	0.1	0	-
U3 - Beta(1, 1)	4.4	1.4	0.4	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						686.3
U2 - Beta(0.5, 0.5)						686
U2 - Beta(0.7, 0.3)						687
U2 - Beta(1, 1)						686.6
U3 - Beta(0.3, 0.7)						685.6
U3 - Beta(0.5, 0.5)						687.2
U3 - Beta(0.7, 0.3)						685
U3 - Beta(1, 1)						686.3

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.9	12.3	18.4	10.5	16.2	32.7
U2 - Beta(0.5, 0.5)	12	13.4	15.3	11.5	15.4	32.4
U2 - Beta(0.7, 0.3)	13.2	12.2	15.3	11	15.2	33.1
U2 - Beta(1, 1)	14.4	12.8	14.5	10.8	14.8	32.7
U3 - Beta(0.3, 0.7)	14.2	14.1	13.6	10.1	15	33
U3 - Beta(0.5, 0.5)	16.1	12.6	13.7	10.7	13.8	33.1
U3 - Beta(0.7, 0.3)	17.8	12.7	12.2	10.7	13.8	32.8
U3 - Beta(1, 1)	20.2	13	11.2	9.7	13.1	32.8
Mean administrations						
U2 - Beta(0.3, 0.7)	7.9	4.4	5.4	3.5	3.5	-
U2 - Beta(0.5, 0.5)	8	4.5	5.2	3.5	3.5	-
U2 - Beta(0.7, 0.3)	8	4.6	5.2	3.5	3.4	-
U2 - Beta(1, 1)	8	4.7	5.1	3.5	3.5	-
U3 - Beta(0.3, 0.7)	7.9	4.5	5.3	3.4	3.5	-
U3 - Beta(0.5, 0.5)	7.9	4.5	5.3	3.5	3.4	-
U3 - Beta(0.7, 0.3)	8	4.7	5.2	3.5	3.4	-
U3 - Beta(1, 1)	8	4.7	5.1	3.4	3.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	2.6	1.9	1.4	1	-
U2 - Beta(0.5, 0.5)	2.1	2.6	1.9	1.4	1	-
U2 - Beta(0.7, 0.3)	2.1	2.6	1.8	1.4	1	-
U2 - Beta(1, 1)	2.1	2.7	1.8	1.4	1	-
U3 - Beta(0.3, 0.7)	2.1	2.6	1.9	1.4	1	-
U3 - Beta(0.5, 0.5)	2.1	2.6	1.9	1.4	1	-
U3 - Beta(0.7, 0.3)	2.1	2.7	1.8	1.4	1	-
U3 - Beta(1, 1)	2.2	2.7	1.8	1.4	1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						861.3
U2 - Beta(0.5, 0.5)						859.2
U2 - Beta(0.7, 0.3)						857.5
U2 - Beta(1, 1)						859.1
U3 - Beta(0.3, 0.7)						859.9
U3 - Beta(0.5, 0.5)						858.5
U3 - Beta(0.7, 0.3)						858.4
U3 - Beta(1, 1)						859

Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.7	25.3	9.2	2	0.6	54.2
U2 - Beta(0.5, 0.5)	8.8	25.7	8.3	1.9	0.7	54.6
U2 - Beta(0.7, 0.3)	8.7	26.9	8.1	1.5	0.6	54.2
U2 - Beta(1, 1)	8.9	25.7	8.9	1.9	0.5	54.1
U3 - Beta(0.3, 0.7)	8.7	26.2	8.4	2	0.5	54.2
U3 - Beta(0.5, 0.5)	9.4	26.4	8	1.4	0.5	54.3
U3 - Beta(0.7, 0.3)	9.4	27.7	6.4	1.8	0.3	54.4
U3 - Beta(1, 1)	9.7	26.7	7.4	1.5	0.4	54.3
Mean administrations						
U2 - Beta(0.3, 0.7)	9.8	5.3	3.4	1.4	0.5	-
U2 - Beta(0.5, 0.5)	9.8	5.4	3.4	1.4	0.4	-
U2 - Beta(0.7, 0.3)	9.8	5.5	3.3	1.4	0.4	-
U2 - Beta(1, 1)	9.8	5.4	3.3	1.4	0.4	-
U3 - Beta(0.3, 0.7)	9.8	5.3	3.4	1.4	0.4	-
U3 - Beta(0.5, 0.5)	9.8	5.4	3.4	1.4	0.4	-
U3 - Beta(0.7, 0.3)	9.8	5.4	3.4	1.4	0.4	-
U3 - Beta(1, 1)	9.8	5.4	3.3	1.4	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.7	2.4	1	0.5	0.1	-
U2 - Beta(0.5, 0.5)	2.7	2.4	0.9	0.5	0.1	-
U2 - Beta(0.7, 0.3)	2.7	2.4	0.9	0.4	0.1	-
U2 - Beta(1, 1)	2.7	2.4	0.9	0.5	0.1	-
U3 - Beta(0.3, 0.7)	2.7	2.4	0.9	0.5	0.1	-
U3 - Beta(0.5, 0.5)	2.7	2.4	0.9	0.4	0.1	-
U3 - Beta(0.7, 0.3)	2.7	2.4	0.9	0.5	0.1	-
U3 - Beta(1, 1)	2.7	2.4	0.9	0.5	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						749.7
U2 - Beta(0.5, 0.5)						748.4
U2 - Beta(0.7, 0.3)						749.2
U2 - Beta(1, 1)						749.5
U3 - Beta(0.3, 0.7)						749.7
U3 - Beta(0.5, 0.5)						748.5
U3 - Beta(0.7, 0.3)						748.6
U3 - Beta(1, 1)						749.1

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	67.9	16.1	1.6	0	0	14.4
U2 - Beta(0.5, 0.5)	66.6	17.4	1.4	0	0	14.6
U2 - Beta(0.7, 0.3)	67.4	17	1.3	0	0	14.3
U2 - Beta(1, 1)	67.7	16.7	1.3	0	0	14.3
U3 - Beta(0.3, 0.7)	68.7	15.8	0.8	0	0	14.7
U3 - Beta(0.5, 0.5)	69.1	15.9	0.9	0	0	14.1
U3 - Beta(0.7, 0.3)	68.7	15.6	1	0	0	14.7
U3 - Beta(1, 1)	69.4	15.7	0.7	0	0	14.2
Mean administrations						
U2 - Beta(0.3, 0.7)	19.1	5.5	2.5	0.5	0	-
U2 - Beta(0.5, 0.5)	19.2	5.4	2.4	0.5	0	-
U2 - Beta(0.7, 0.3)	19.2	5.4	2.4	0.5	0	-
U2 - Beta(1, 1)	19.2	5.5	2.4	0.5	0	-
U3 - Beta(0.3, 0.7)	19.1	5.5	2.4	0.5	0	-
U3 - Beta(0.5, 0.5)	19.2	5.4	2.4	0.5	0	-
U3 - Beta(0.7, 0.3)	19.2	5.4	2.4	0.5	0	-
U3 - Beta(1, 1)	19.3	5.4	2.4	0.5	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	6	2.5	0.8	0.2	0	-
U2 - Beta(0.5, 0.5)	6	2.5	0.8	0.2	0	-
U2 - Beta(0.7, 0.3)	6	2.4	0.8	0.2	0	-
U2 - Beta(1, 1)	6	2.4	0.8	0.2	0	-
U3 - Beta(0.3, 0.7)	6	2.4	0.8	0.2	0	-
U3 - Beta(0.5, 0.5)	6	2.5	0.8	0.2	0	-
U3 - Beta(0.7, 0.3)	6	2.4	0.8	0.2	0	-
U3 - Beta(1, 1)	6.1	2.4	0.8	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						994.6
U2 - Beta(0.5, 0.5)						994.1
U2 - Beta(0.7, 0.3)						994.4
U2 - Beta(1, 1)						993.8
U3 - Beta(0.3, 0.7)						993
U3 - Beta(0.5, 0.5)						994.6
U3 - Beta(0.7, 0.3)						992.9
U3 - Beta(1, 1)						993.9

Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.8	16.5	13.4	37.4	13.1	5.8
U2 - Beta(0.5, 0.5)	15	18	11.2	37.1	12.9	5.8
U2 - Beta(0.7, 0.3)	16.9	15.9	12.2	36.8	12.6	5.6
U2 - Beta(1, 1)	17.6	16	10	37.3	13.2	5.9
U3 - Beta(0.3, 0.7)	18.7	17.7	10.5	34	13.1	6
U3 - Beta(0.5, 0.5)	22.2	16.3	8.7	35.4	11.9	5.5
U3 - Beta(0.7, 0.3)	23.6	17.8	7.3	33.9	11.8	5.6
U3 - Beta(1, 1)	25.2	17.1	7	33.3	11.9	5.5
Mean administrations						
U2 - Beta(0.3, 0.7)	4.9	5.1	6	7.6	5.5	-
U2 - Beta(0.5, 0.5)	5	5.2	6	7.4	5.5	-
U2 - Beta(0.7, 0.3)	5	5.4	5.9	7.3	5.4	-
U2 - Beta(1, 1)	5.1	5.4	5.8	7.2	5.6	-
U3 - Beta(0.3, 0.7)	4.9	5.1	6	7.5	5.5	-
U3 - Beta(0.5, 0.5)	4.9	5.3	6	7.4	5.5	-
U3 - Beta(0.7, 0.3)	5.1	5.5	5.8	7.2	5.4	-
U3 - Beta(1, 1)	5.1	5.4	5.8	7.3	5.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	3.1	1.9	2.8	1.3	-
U2 - Beta(0.5, 0.5)	0.9	3.2	1.9	2.8	1.3	-
U2 - Beta(0.7, 0.3)	0.9	3.2	1.9	2.7	1.3	-
U2 - Beta(1, 1)	1	3.2	1.9	2.7	1.3	-
U3 - Beta(0.3, 0.7)	0.9	3.1	1.9	2.8	1.3	-
U3 - Beta(0.5, 0.5)	0.9	3.2	1.9	2.8	1.3	-
U3 - Beta(0.7, 0.3)	1	3.2	1.9	2.7	1.2	-
U3 - Beta(1, 1)	1	3.2	1.9	2.8	1.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1048.6
U2 - Beta(0.5, 0.5)						1047.4
U2 - Beta(0.7, 0.3)						1045.5
U2 - Beta(1, 1)						1046.6
U3 - Beta(0.3, 0.7)						1048.3
U3 - Beta(0.5, 0.5)						1047.4
U3 - Beta(0.7, 0.3)						1046.2
U3 - Beta(1, 1)						1046.7

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.5	26.5	10	2	0.6	50.4
U2 - Beta(0.5, 0.5)	10.6	27	9.6	1.9	0.6	50.3
U2 - Beta(0.7, 0.3)	10.9	27.5	9.1	1.8	0.7	50
U2 - Beta(1, 1)	11.2	27.2	9.2	1.8	0.5	50.1
U3 - Beta(0.3, 0.7)	11.7	26.8	8.8	1.9	0.6	50.2
U3 - Beta(0.5, 0.5)	11.8	26	9.4	1.8	0.6	50.4
U3 - Beta(0.7, 0.3)	12	27.7	7.6	1.7	0.5	50.5
U3 - Beta(1, 1)	12.4	27.1	8.4	1.4	0.5	50.2
Mean administrations						
U2 - Beta(0.3, 0.7)	9.6	5.7	3.7	1.5	0.5	-
U2 - Beta(0.5, 0.5)	9.6	5.8	3.6	1.5	0.5	-
U2 - Beta(0.7, 0.3)	9.6	5.9	3.6	1.4	0.6	-
U2 - Beta(1, 1)	9.6	5.9	3.6	1.5	0.5	-
U3 - Beta(0.3, 0.7)	9.6	5.7	3.8	1.5	0.5	-
U3 - Beta(0.5, 0.5)	9.6	5.8	3.7	1.4	0.5	-
U3 - Beta(0.7, 0.3)	9.6	5.8	3.6	1.4	0.5	-
U3 - Beta(1, 1)	9.6	5.8	3.7	1.4	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.7	2.7	1.1	0.6	0.1	-
U2 - Beta(0.5, 0.5)	2.7	2.8	1.1	0.6	0.1	-
U2 - Beta(0.7, 0.3)	2.7	2.8	1.1	0.5	0.2	-
U2 - Beta(1, 1)	2.7	2.8	1.1	0.5	0.2	-
U3 - Beta(0.3, 0.7)	2.7	2.7	1.1	0.5	0.2	-
U3 - Beta(0.5, 0.5)	2.7	2.7	1.1	0.5	0.2	-
U3 - Beta(0.7, 0.3)	2.7	2.8	1.1	0.5	0.1	-
U3 - Beta(1, 1)	2.7	2.7	1.1	0.5	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						756.8
U2 - Beta(0.5, 0.5)						756.8
U2 - Beta(0.7, 0.3)						757.9
U2 - Beta(1, 1)						756.7
U3 - Beta(0.3, 0.7)						756.1
U3 - Beta(0.5, 0.5)						755.9
U3 - Beta(0.7, 0.3)						755.5
U3 - Beta(1, 1)						757.3

Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	15.7	17.7	16.3	23.5	10.1	16.7
U2 - Beta(0.5, 0.5)	18.7	16.2	16.5	22.8	9.3	16.5
U2 - Beta(0.7, 0.3)	20.1	16.6	14.3	23.1	9.2	16.7
U2 - Beta(1, 1)	19.7	17.7	13.4	23.3	9	16.9
U3 - Beta(0.3, 0.7)	22.8	16.7	12.9	22.2	8.4	17
U3 - Beta(0.5, 0.5)	27.2	16.7	10.9	21	7.5	16.7
U3 - Beta(0.7, 0.3)	28.5	16.7	10.3	20.5	6.9	17.1
U3 - Beta(1, 1)	30.7	16.2	9.4	20.5	6.4	16.8
Mean administrations						
U2 - Beta(0.3, 0.7)	6.6	5.6	6.2	5.2	3.9	-
U2 - Beta(0.5, 0.5)	6.7	5.5	6	5.3	3.9	-
U2 - Beta(0.7, 0.3)	6.8	5.8	5.8	5.2	3.8	-
U2 - Beta(1, 1)	6.8	5.8	5.8	5.2	3.8	-
U3 - Beta(0.3, 0.7)	6.5	5.6	6.2	5.4	3.8	-
U3 - Beta(0.5, 0.5)	6.7	5.7	6	5.3	3.8	-
U3 - Beta(0.7, 0.3)	6.7	5.8	5.9	5.3	3.7	-
U3 - Beta(1, 1)	6.7	5.8	5.9	5.3	3.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.6	3.2	2.2	2	1.2	-
U2 - Beta(0.5, 0.5)	1.7	3.2	2.1	2.1	1.2	-
U2 - Beta(0.7, 0.3)	1.7	3.3	2	2.1	1.2	-
U2 - Beta(1, 1)	1.7	3.3	2.1	2.1	1.2	-
U3 - Beta(0.3, 0.7)	1.6	3.2	2.2	2.1	1.2	-
U3 - Beta(0.5, 0.5)	1.6	3.2	2.1	2.1	1.2	-
U3 - Beta(0.7, 0.3)	1.6	3.3	2.1	2.1	1.1	-
U3 - Beta(1, 1)	1.7	3.3	2	2.1	1.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						948.4
U2 - Beta(0.5, 0.5)						948.3
U2 - Beta(0.7, 0.3)						944.7
U2 - Beta(1, 1)						944
U3 - Beta(0.3, 0.7)						948.9
U3 - Beta(0.5, 0.5)						945.9
U3 - Beta(0.7, 0.3)						945.1
U3 - Beta(1, 1)						946.4

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	40.8	29.2	8	2.9	0.1	19
U2 - Beta(0.5, 0.5)	42.9	26.2	9.3	2.7	0.1	18.8
U2 - Beta(0.7, 0.3)	43.1	25	9.7	3	0.1	19.1
U2 - Beta(1, 1)	45.5	25.2	8.2	1.8	0.2	19.1
U3 - Beta(0.3, 0.7)	48.1	25.3	6	1.2	0.2	19.2
U3 - Beta(0.5, 0.5)	49.1	25.8	4.8	1.1	0.1	19.1
U3 - Beta(0.7, 0.3)	52.1	22.6	4.5	1.2	0.2	19.4
U3 - Beta(1, 1)	55.5	20.3	4.2	1.1	0.1	18.8
Mean administrations						
U2 - Beta(0.3, 0.7)	11.2	7.3	5.5	2.5	0.9	-
U2 - Beta(0.5, 0.5)	11.3	7.3	5.4	2.4	0.9	-
U2 - Beta(0.7, 0.3)	11.3	7.5	5.3	2.5	0.8	-
U2 - Beta(1, 1)	11.4	7.4	5.3	2.4	0.8	-
U3 - Beta(0.3, 0.7)	11.1	7.3	5.6	2.4	0.9	-
U3 - Beta(0.5, 0.5)	11.3	7.4	5.4	2.4	0.8	-
U3 - Beta(0.7, 0.3)	11.2	7.5	5.4	2.4	0.9	-
U3 - Beta(1, 1)	11.4	7.4	5.4	2.4	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.5	3.7	1.9	1.1	0.3	-
U2 - Beta(0.5, 0.5)	3.5	3.7	1.9	1.1	0.4	-
U2 - Beta(0.7, 0.3)	3.5	3.8	1.9	1.1	0.3	-
U2 - Beta(1, 1)	3.5	3.7	1.9	1.1	0.4	-
U3 - Beta(0.3, 0.7)	3.4	3.7	2	1.1	0.3	-
U3 - Beta(0.5, 0.5)	3.5	3.7	1.9	1.1	0.3	-
U3 - Beta(0.7, 0.3)	3.4	3.8	1.9	1.1	0.4	-
U3 - Beta(1, 1)	3.5	3.7	1.9	1.1	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						923.1
U2 - Beta(0.5, 0.5)						923.5
U2 - Beta(0.7, 0.3)						922.7
U2 - Beta(1, 1)						922.8
U3 - Beta(0.3, 0.7)						922.9
U3 - Beta(0.5, 0.5)						923.2
U3 - Beta(0.7, 0.3)						923
U3 - Beta(1, 1)						923.4
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	27.6	27.5	17.2	16.4	11.2	0.1
U2 - Beta(0.5, 0.5)	27	27.2	16.6	17	12.1	0.1
U2 - Beta(0.7, 0.3)	28.2	30	15.3	15.9	10.5	0.1
U2 - Beta(1, 1)	28.4	30.1	16.2	14.5	10.7	0.1
U3 - Beta(0.3, 0.7)	33.2	32	14.5	13.7	6.5	0.1
U3 - Beta(0.5, 0.5)	34.1	32.7	14.1	12	7	0.1
U3 - Beta(0.7, 0.3)	34.6	33.7	14.4	11	6.2	0.1
U3 - Beta(1, 1)	37.5	34.2	13.4	10.4	4.4	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	5.7	5.7	5.6	5.8	7.3	-
U2 - Beta(0.5, 0.5)	5.7	5.9	5.5	5.8	7	-
U2 - Beta(0.7, 0.3)	5.7	5.9	5.5	6	6.9	-
U2 - Beta(1, 1)	5.7	5.9	5.6	5.9	6.9	-
U3 - Beta(0.3, 0.7)	5.7	5.6	5.5	5.9	7.2	-
U3 - Beta(0.5, 0.5)	5.7	5.9	5.5	5.8	7	-
U3 - Beta(0.7, 0.3)	5.7	6	5.5	5.9	6.9	-
U3 - Beta(1, 1)	5.7	5.9	5.5	5.9	6.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.1	3.2	1.9	2.6	2.8	-
U2 - Beta(0.5, 0.5)	1.1	3.3	1.9	2.6	2.7	-
U2 - Beta(0.7, 0.3)	1.1	3.3	1.9	2.7	2.6	-
U2 - Beta(1, 1)	1.1	3.3	1.9	2.7	2.6	-
U3 - Beta(0.3, 0.7)	1.1	3.2	1.9	2.7	2.8	-
U3 - Beta(0.5, 0.5)	1.1	3.3	1.9	2.6	2.7	-
U3 - Beta(0.7, 0.3)	1.1	3.4	1.9	2.7	2.6	-
U3 - Beta(1, 1)	1.1	3.3	1.9	2.7	2.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1008.1
U2 - Beta(0.5, 0.5)						1006.6
U2 - Beta(0.7, 0.3)						1006.9
U2 - Beta(1, 1)						1006.6
U3 - Beta(0.3, 0.7)						1008.3
U3 - Beta(0.5, 0.5)						1006.8
U3 - Beta(0.7, 0.3)						1006.9
U3 - Beta(1, 1)						1006.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	32.1	53.2	13.5	0.7	0	0.5
U2 - Beta(0.5, 0.5)	32.8	53.9	11.9	0.8	0	0.6
U2 - Beta(0.7, 0.3)	31.4	54.4	12.3	1.4	0	0.5
U2 - Beta(1, 1)	32.4	53.9	12.2	1	0	0.5
U3 - Beta(0.3, 0.7)	34.7	54	10.2	0.6	0	0.5
U3 - Beta(0.5, 0.5)	35.2	53.6	10.1	0.6	0	0.5
U3 - Beta(0.7, 0.3)	33.7	55.2	9.9	0.8	0	0.4
U3 - Beta(1, 1)	34.6	55.4	8.9	0.6	0	0.5
Mean administrations						
U2 - Beta(0.3, 0.7)	11.1	10.5	6.6	1.5	0.2	-
U2 - Beta(0.5, 0.5)	11	10.6	6.5	1.5	0.2	-
U2 - Beta(0.7, 0.3)	11.2	10.5	6.4	1.5	0.2	-
U2 - Beta(1, 1)	11	10.6	6.6	1.5	0.2	-
U3 - Beta(0.3, 0.7)	11	10.5	6.6	1.6	0.2	-
U3 - Beta(0.5, 0.5)	11.1	10.5	6.5	1.5	0.2	-
U3 - Beta(0.7, 0.3)	11.1	10.7	6.3	1.5	0.2	-
U3 - Beta(1, 1)	11.1	10.5	6.5	1.5	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	3.6	1.5	0.5	0.1	-
U2 - Beta(0.5, 0.5)	2.8	3.6	1.5	0.4	0.1	-
U2 - Beta(0.7, 0.3)	2.9	3.7	1.5	0.4	0.1	-
U2 - Beta(1, 1)	2.8	3.6	1.5	0.5	0.1	-
U3 - Beta(0.3, 0.7)	2.8	3.6	1.5	0.5	0.1	-
U3 - Beta(0.5, 0.5)	2.9	3.6	1.5	0.5	0	-
U3 - Beta(0.7, 0.3)	2.8	3.7	1.4	0.5	0	-
U3 - Beta(1, 1)	2.9	3.6	1.5	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1184
U2 - Beta(0.5, 0.5)						1183.9
U2 - Beta(0.7, 0.3)						1182.3
U2 - Beta(1, 1)						1183.9
U3 - Beta(0.3, 0.7)						1184.7
U3 - Beta(0.5, 0.5)						1183.5
U3 - Beta(0.7, 0.3)						1183
U3 - Beta(1, 1)						1184.4

Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.7	11.6	12.1	18.6	49.1	0.9
U2 - Beta(0.5, 0.5)	9.5	12.4	11.2	17.5	48.5	0.9
U2 - Beta(0.7, 0.3)	9	13.9	11	17.5	47.7	0.9
U2 - Beta(1, 1)	11.2	13.1	11.3	16.8	46.7	0.9
U3 - Beta(0.3, 0.7)	13.9	14.2	11.1	15.7	44.3	0.8
U3 - Beta(0.5, 0.5)	15.4	15.2	11	15.4	42	1
U3 - Beta(0.7, 0.3)	15.7	15.7	11.4	15.5	40.7	1
U3 - Beta(1, 1)	19	15.5	11	13.3	40.3	0.9
Mean administrations						
U2 - Beta(0.3, 0.7)	3.7	3.9	4.8	5.7	11.8	-
U2 - Beta(0.5, 0.5)	3.7	4	4.9	5.7	11.6	-
U2 - Beta(0.7, 0.3)	3.8	4.1	4.9	5.8	11.4	-
U2 - Beta(1, 1)	3.7	4.1	4.9	5.6	11.5	-
U3 - Beta(0.3, 0.7)	3.7	3.9	4.8	5.8	11.7	-
U3 - Beta(0.5, 0.5)	3.7	4	4.8	5.8	11.6	-
U3 - Beta(0.7, 0.3)	3.7	4.1	4.9	5.7	11.4	-
U3 - Beta(1, 1)	3.8	4	4.9	5.7	11.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.3	2.7	1.6	2.5	3.3	-
U2 - Beta(0.5, 0.5)	0.3	2.8	1.6	2.5	3.2	-
U2 - Beta(0.7, 0.3)	0.4	2.8	1.6	2.6	3.2	-
U2 - Beta(1, 1)	0.3	2.8	1.6	2.5	3.2	-
U3 - Beta(0.3, 0.7)	0.3	2.7	1.6	2.6	3.2	-
U3 - Beta(0.5, 0.5)	0.4	2.8	1.6	2.6	3.2	-
U3 - Beta(0.7, 0.3)	0.4	2.8	1.6	2.6	3.2	-
U3 - Beta(1, 1)	0.4	2.8	1.6	2.5	3.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1068.6
U2 - Beta(0.5, 0.5)						1065.6
U2 - Beta(0.7, 0.3)						1063.9
U2 - Beta(1, 1)						1065.4
U3 - Beta(0.3, 0.7)						1068
U3 - Beta(0.5, 0.5)						1065.1
U3 - Beta(0.7, 0.3)						1064.1
U3 - Beta(1, 1)						1064.3

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	22.2	48.5	11.3	0.8	0	17.2
U2 - Beta(0.5, 0.5)	22	49	10.7	0.7	0	17.6
U2 - Beta(0.7, 0.3)	23.5	48.2	10.4	0.5	0	17.4
U2 - Beta(1, 1)	23.1	49	9.9	0.7	0	17.3
U3 - Beta(0.3, 0.7)	24.2	49.6	8.8	0.2	0	17.2
U3 - Beta(0.5, 0.5)	23.8	48.6	9.1	0.7	0	17.8
U3 - Beta(0.7, 0.3)	25.5	47.4	9.2	0.6	0	17.3
U3 - Beta(1, 1)	26.3	47.5	8.4	0.5	0	17.3
Mean administrations						
U2 - Beta(0.3, 0.7)	8.9	10.5	6.2	1.3	0.2	-
U2 - Beta(0.5, 0.5)	9	10.5	6	1.3	0.2	-
U2 - Beta(0.7, 0.3)	9	10.7	5.9	1.3	0.2	-
U2 - Beta(1, 1)	9	10.6	5.9	1.3	0.2	-
U3 - Beta(0.3, 0.7)	8.9	10.6	6.1	1.3	0.2	-
U3 - Beta(0.5, 0.5)	9	10.6	6.1	1.3	0.2	-
U3 - Beta(0.7, 0.3)	9	10.7	5.9	1.3	0.2	-
U3 - Beta(1, 1)	9	10.7	5.8	1.3	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	4.5	1.8	0.5	0	-
U2 - Beta(0.5, 0.5)	2.5	4.6	1.8	0.5	0	-
U2 - Beta(0.7, 0.3)	2.5	4.6	1.7	0.5	0	-
U2 - Beta(1, 1)	2.5	4.6	1.7	0.5	0	-
U3 - Beta(0.3, 0.7)	2.4	4.6	1.8	0.5	0	-
U3 - Beta(0.5, 0.5)	2.5	4.5	1.8	0.5	0	-
U3 - Beta(0.7, 0.3)	2.5	4.6	1.7	0.5	0	-
U3 - Beta(1, 1)	2.5	4.6	1.7	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						976.3
U2 - Beta(0.5, 0.5)						976.3
U2 - Beta(0.7, 0.3)						975.7
U2 - Beta(1, 1)						974.5
U3 - Beta(0.3, 0.7)						976.8
U3 - Beta(0.5, 0.5)						976.1
U3 - Beta(0.7, 0.3)						974.8
U3 - Beta(1, 1)						975.8

Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.2	48.1	27.4	6.2	0.2	7.9
U2 - Beta(0.5, 0.5)	10.6	47.7	27.7	5.9	0.3	7.8
U2 - Beta(0.7, 0.3)	11.5	48.1	26.1	6.3	0.2	7.8
U2 - Beta(1, 1)	11.3	49.2	25.4	6	0.3	7.8
U3 - Beta(0.3, 0.7)	14.2	50.8	23.2	4.1	0	7.7
U3 - Beta(0.5, 0.5)	14.3	51.2	22.3	3.9	0.4	7.9
U3 - Beta(0.7, 0.3)	16.3	50.1	20.7	4.8	0.3	7.8
U3 - Beta(1, 1)	19.6	50.4	17.5	4.6	0.2	7.7
Mean administrations						
U2 - Beta(0.3, 0.7)	5.4	9.4	8.6	3.7	1.4	-
U2 - Beta(0.5, 0.5)	5.4	9.5	8.5	3.7	1.4	-
U2 - Beta(0.7, 0.3)	5.5	9.6	8.4	3.7	1.4	-
U2 - Beta(1, 1)	5.5	9.7	8.1	3.8	1.4	-
U3 - Beta(0.3, 0.7)	5.4	9.4	8.6	3.7	1.4	-
U3 - Beta(0.5, 0.5)	5.4	9.5	8.5	3.7	1.4	-
U3 - Beta(0.7, 0.3)	5.4	9.7	8.3	3.7	1.3	-
U3 - Beta(1, 1)	5.5	9.7	8.3	3.7	1.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1	4.6	2.7	1.4	0.5	-
U2 - Beta(0.5, 0.5)	1.1	4.6	2.6	1.4	0.5	-
U2 - Beta(0.7, 0.3)	1.1	4.6	2.6	1.4	0.5	-
U2 - Beta(1, 1)	1.1	4.7	2.5	1.5	0.5	-
U3 - Beta(0.3, 0.7)	1.1	4.5	2.7	1.4	0.5	-
U3 - Beta(0.5, 0.5)	1.1	4.6	2.6	1.4	0.5	-
U3 - Beta(0.7, 0.3)	1.1	4.7	2.6	1.4	0.5	-
U3 - Beta(1, 1)	1.1	4.7	2.6	1.4	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1005.6
U2 - Beta(0.5, 0.5)						1004.8
U2 - Beta(0.7, 0.3)						1004.1
U2 - Beta(1, 1)						1004.7
U3 - Beta(0.3, 0.7)						1005.6
U3 - Beta(0.5, 0.5)						1004.6
U3 - Beta(0.7, 0.3)						1004.1
U3 - Beta(1, 1)						1005.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	47.3	0.3	0	0	0	52.4
U2 - Beta(0.5, 0.5)	47.3	0.5	0	0	0	52.2
U2 - Beta(0.7, 0.3)	46.8	0.6	0	0	0	52.6
U2 - Beta(1, 1)	47.1	0.5	0	0	0	52.4
U3 - Beta(0.3, 0.7)	47.4	0.5	0	0	0	52.1
U3 - Beta(0.5, 0.5)	47.2	0.5	0	0	0	52.3
U3 - Beta(0.7, 0.3)	47	0.5	0	0	0	52.5
U3 - Beta(1, 1)	47.7	0.5	0	0	0	51.8
Mean administrations						
U2 - Beta(0.3, 0.7)	19.7	1.6	0.3	0	0	-
U2 - Beta(0.5, 0.5)	19.7	1.6	0.3	0	0	-
U2 - Beta(0.7, 0.3)	19.7	1.6	0.3	0	0	-
U2 - Beta(1, 1)	19.7	1.6	0.3	0	0	-
U3 - Beta(0.3, 0.7)	19.7	1.6	0.3	0	0	-
U3 - Beta(0.5, 0.5)	19.7	1.6	0.3	0	0	-
U3 - Beta(0.7, 0.3)	19.7	1.6	0.3	0	0	-
U3 - Beta(1, 1)	19.7	1.6	0.3	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4	0.5	0.1	0	0	-
U2 - Beta(0.5, 0.5)	4	0.5	0.1	0	0	-
U2 - Beta(0.7, 0.3)	4	0.5	0.1	0	0	-
U2 - Beta(1, 1)	4	0.5	0.1	0	0	-
U3 - Beta(0.3, 0.7)	4	0.5	0.1	0	0	-
U3 - Beta(0.5, 0.5)	4	0.5	0.1	0	0	-
U3 - Beta(0.7, 0.3)	4	0.5	0.1	0	0	-
U3 - Beta(1, 1)	4	0.5	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						934.9
U2 - Beta(0.5, 0.5)						935
U2 - Beta(0.7, 0.3)						933.1
U2 - Beta(1, 1)						934.3
U3 - Beta(0.3, 0.7)						934.9
U3 - Beta(0.5, 0.5)						933.6
U3 - Beta(0.7, 0.3)						934.4
U3 - Beta(1, 1)						935.7
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	26.2	55.3	6.9	0.6	0	11
U2 - Beta(0.5, 0.5)	27.4	53.3	7.8	0.5	0	11
U2 - Beta(0.7, 0.3)	26.8	53.5	8.2	0.5	0	11
U2 - Beta(1, 1)	26.2	53.8	8.7	0.5	0	10.8
U3 - Beta(0.3, 0.7)	29	53.7	5.7	0.7	0	10.9
U3 - Beta(0.5, 0.5)	28.4	53.6	6.3	0.6	0	11.1
U3 - Beta(0.7, 0.3)	29.3	52	6.9	0.4	0	11.4
U3 - Beta(1, 1)	29.7	52	6.5	0.5	0	11.3
Mean administrations						
U2 - Beta(0.3, 0.7)	8.7	13.1	5.3	0.9	0.1	-
U2 - Beta(0.5, 0.5)	8.8	13	5.2	1	0.1	-
U2 - Beta(0.7, 0.3)	8.8	12.9	5.2	0.9	0.1	-
U2 - Beta(1, 1)	8.9	12.8	5.2	1	0.1	-
U3 - Beta(0.3, 0.7)	8.6	13.1	5.2	0.9	0.1	-
U3 - Beta(0.5, 0.5)	8.8	13	5.2	1	0.1	-
U3 - Beta(0.7, 0.3)	8.8	12.9	5.2	1	0.1	-
U3 - Beta(1, 1)	8.9	12.9	5.2	1	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.3	5.6	1.6	0.4	0	-
U2 - Beta(0.5, 0.5)	2.3	5.6	1.5	0.4	0	-
U2 - Beta(0.7, 0.3)	2.3	5.6	1.5	0.4	0	-
U2 - Beta(1, 1)	2.4	5.6	1.5	0.4	0	-
U3 - Beta(0.3, 0.7)	2.3	5.6	1.5	0.4	0	-
U3 - Beta(0.5, 0.5)	2.3	5.6	1.5	0.4	0	-
U3 - Beta(0.7, 0.3)	2.3	5.6	1.5	0.4	0	-
U3 - Beta(1, 1)	2.4	5.6	1.5	0.4	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						994.9
U2 - Beta(0.5, 0.5)						994.1
U2 - Beta(0.7, 0.3)						993.9
U2 - Beta(1, 1)						993.2
U3 - Beta(0.3, 0.7)						994.6
U3 - Beta(0.5, 0.5)						992.5
U3 - Beta(0.7, 0.3)						993.6
U3 - Beta(1, 1)						993.3

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6	1.6	0	0	0	92.4
U2 - Beta(0.5, 0.5)	6.1	1.6	0	0	0	92.3
U2 - Beta(0.7, 0.3)	6.1	1.8	0	0	0	92.1
U2 - Beta(1, 1)	6.2	1.8	0	0	0	92
U3 - Beta(0.3, 0.7)	6.1	1.5	0	0	0	92.4
U3 - Beta(0.5, 0.5)	6.1	1.5	0	0	0	92.4
U3 - Beta(0.7, 0.3)	6.4	1.6	0	0	0	92
U3 - Beta(1, 1)	6.5	1.6	0	0	0	91.9
Mean administrations						
U2 - Beta(0.3, 0.7)	11.8	1.7	0.2	0	0	-
U2 - Beta(0.5, 0.5)	11.7	1.7	0.2	0	0	-
U2 - Beta(0.7, 0.3)	11.7	1.7	0.2	0	0	-
U2 - Beta(1, 1)	11.7	1.7	0.3	0	0	-
U3 - Beta(0.3, 0.7)	11.7	1.7	0.2	0	0	-
U3 - Beta(0.5, 0.5)	11.7	1.7	0.2	0	0	-
U3 - Beta(0.7, 0.3)	11.8	1.7	0.2	0	0	-
U3 - Beta(1, 1)	11.8	1.7	0.2	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.4	1	0.1	0	0	-
U2 - Beta(0.5, 0.5)	3.4	1	0.1	0	0	-
U2 - Beta(0.7, 0.3)	3.4	1	0.1	0	0	-
U2 - Beta(1, 1)	3.4	1	0.1	0	0	-
U3 - Beta(0.3, 0.7)	3.4	1	0.1	0	0	-
U3 - Beta(0.5, 0.5)	3.4	1	0.1	0	0	-
U3 - Beta(0.7, 0.3)	3.4	1	0.1	0	0	-
U3 - Beta(1, 1)	3.4	1	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						498.2
U2 - Beta(0.5, 0.5)						498.4
U2 - Beta(0.7, 0.3)						498.2
U2 - Beta(1, 1)						498.9
U3 - Beta(0.3, 0.7)						497.5
U3 - Beta(0.5, 0.5)						497.9
U3 - Beta(0.7, 0.3)						499.1
U3 - Beta(1, 1)						498.1
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	0.4	1.2	4.3	46.4	47.6	0.1
U2 - Beta(0.5, 0.5)	0.7	1.7	4	49.9	43.6	0.1
U2 - Beta(0.7, 0.3)	0.6	1.8	4.3	49.8	43.3	0.2
U2 - Beta(1, 1)	1.3	1.6	3.9	49.6	43.4	0.2
U3 - Beta(0.3, 0.7)	1.7	2.2	5	47.3	43.7	0.1
U3 - Beta(0.5, 0.5)	1.8	2.6	4.8	51.1	39.6	0.1
U3 - Beta(0.7, 0.3)	2	3.5	5.6	52.2	36.6	0.1
U3 - Beta(1, 1)	2.1	3.3	5.7	50.7	38.1	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.2	3.1	3.6	8.1	11.9	-
U2 - Beta(0.5, 0.5)	3.2	3.1	3.6	8.4	11.7	-
U2 - Beta(0.7, 0.3)	3.2	3.2	3.7	8.6	11.3	-
U2 - Beta(1, 1)	3.2	3.1	3.6	8.4	11.6	-
U3 - Beta(0.3, 0.7)	3.2	3.1	3.6	8.1	11.9	-
U3 - Beta(0.5, 0.5)	3.2	3.1	3.6	8.5	11.5	-
U3 - Beta(0.7, 0.3)	3.2	3.2	3.7	8.6	11.3	-
U3 - Beta(1, 1)	3.2	3.1	3.6	8.3	11.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.1	2.4	1.1	3.2	3.2	-
U2 - Beta(0.5, 0.5)	0.1	2.4	1.1	3.2	3.2	-
U2 - Beta(0.7, 0.3)	0.1	2.4	1.2	3.3	3.1	-
U2 - Beta(1, 1)	0.1	2.4	1.1	3.3	3.1	-
U3 - Beta(0.3, 0.7)	0.1	2.4	1.1	3.2	3.2	-
U3 - Beta(0.5, 0.5)	0.1	2.4	1.1	3.3	3.1	-
U3 - Beta(0.7, 0.3)	0.1	2.4	1.2	3.3	3.1	-
U3 - Beta(1, 1)	0.1	2.4	1.1	3.2	3.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1102
U2 - Beta(0.5, 0.5)						1101.3
U2 - Beta(0.7, 0.3)						1099
U2 - Beta(1, 1)						1100.7
U3 - Beta(0.3, 0.7)						1102.1
U3 - Beta(0.5, 0.5)						1101
U3 - Beta(0.7, 0.3)						1099.3
U3 - Beta(1, 1)						1101.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	36.5	33.9	24.9	2.7	0	2
U2 - Beta(0.5, 0.5)	35.4	35.2	24.2	3.2	0	2
U2 - Beta(0.7, 0.3)	34.9	36	23.6	3.4	0	2.1
U2 - Beta(1, 1)	36.3	35.4	23.5	2.7	0	2.1
U3 - Beta(0.3, 0.7)	39.9	34.1	21.5	2.4	0	2.1
U3 - Beta(0.5, 0.5)	40.6	33.6	21.7	2	0	2.1
U3 - Beta(0.7, 0.3)	40.5	35.2	19.1	3.1	0	2.1
U3 - Beta(1, 1)	41.5	34.3	19.5	2.6	0	2.1
Mean administrations						
U2 - Beta(0.3, 0.7)	10	8.6	8.9	1.9	0.2	-
U2 - Beta(0.5, 0.5)	10.1	8.8	8.5	2	0.2	-
U2 - Beta(0.7, 0.3)	10.1	8.9	8.4	1.9	0.3	-
U2 - Beta(1, 1)	10.2	8.8	8.5	1.9	0.2	-
U3 - Beta(0.3, 0.7)	10	8.7	8.8	1.9	0.2	-
U3 - Beta(0.5, 0.5)	10.1	8.8	8.5	1.9	0.2	-
U3 - Beta(0.7, 0.3)	10.1	9	8.3	2	0.2	-
U3 - Beta(1, 1)	10.2	8.8	8.5	1.9	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	3.8	2.4	0.7	0.1	-
U2 - Beta(0.5, 0.5)	2.8	3.9	2.4	0.7	0.1	-
U2 - Beta(0.7, 0.3)	2.9	3.9	2.3	0.7	0.1	-
U2 - Beta(1, 1)	2.9	3.9	2.3	0.7	0.1	-
U3 - Beta(0.3, 0.7)	2.8	3.8	2.4	0.7	0.1	-
U3 - Beta(0.5, 0.5)	2.8	3.9	2.3	0.7	0.1	-
U3 - Beta(0.7, 0.3)	2.8	3.9	2.3	0.7	0.1	-
U3 - Beta(1, 1)	2.9	3.9	2.3	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1097.2
U2 - Beta(0.5, 0.5)						1095.1
U2 - Beta(0.7, 0.3)						1093
U2 - Beta(1, 1)						1094.4
U3 - Beta(0.3, 0.7)						1097.2
U3 - Beta(0.5, 0.5)						1094.5
U3 - Beta(0.7, 0.3)						1094.1
U3 - Beta(1, 1)						1094.9

Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	28.2	28.6	20.9	2.2	0	20.1
U2 - Beta(0.5, 0.5)	27.9	29	20.5	2.7	0	19.9
U2 - Beta(0.7, 0.3)	28.9	28.4	19.7	2.8	0	20.2
U2 - Beta(1, 1)	28	29.7	19.5	2.7	0	20.1
U3 - Beta(0.3, 0.7)	30	29.1	18.8	2.2	0	19.9
U3 - Beta(0.5, 0.5)	30.6	29.1	18.5	1.9	0	19.9
U3 - Beta(0.7, 0.3)	31.7	28.6	17.3	2.3	0	20.1
U3 - Beta(1, 1)	32.3	29.2	16	2.5	0	20
Mean administrations						
U2 - Beta(0.3, 0.7)	10.8	7.1	6.9	1.9	0.2	-
U2 - Beta(0.5, 0.5)	10.8	7.3	6.8	1.9	0.2	-
U2 - Beta(0.7, 0.3)	10.8	7.3	6.7	1.8	0.2	-
U2 - Beta(1, 1)	10.8	7.3	6.7	1.8	0.2	-
U3 - Beta(0.3, 0.7)	10.8	7.1	6.9	1.9	0.2	-
U3 - Beta(0.5, 0.5)	10.7	7.3	6.7	1.9	0.2	-
U3 - Beta(0.7, 0.3)	10.8	7.3	6.7	1.8	0.2	-
U3 - Beta(1, 1)	10.8	7.3	6.7	1.9	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	3.4	2	0.7	0.1	-
U2 - Beta(0.5, 0.5)	3.2	3.4	2	0.7	0.1	-
U2 - Beta(0.7, 0.3)	3.2	3.4	2	0.6	0.1	-
U2 - Beta(1, 1)	3.2	3.4	2	0.7	0	-
U3 - Beta(0.3, 0.7)	3.2	3.4	2.1	0.7	0.1	-
U3 - Beta(0.5, 0.5)	3.2	3.5	2	0.7	0.1	-
U3 - Beta(0.7, 0.3)	3.2	3.4	2	0.7	0.1	-
U3 - Beta(1, 1)	3.2	3.5	2	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						964.2
U2 - Beta(0.5, 0.5)						964.7
U2 - Beta(0.7, 0.3)						962.5
U2 - Beta(1, 1)						963
U3 - Beta(0.3, 0.7)						964.9
U3 - Beta(0.5, 0.5)						964
U3 - Beta(0.7, 0.3)						963.5
U3 - Beta(1, 1)						964.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	19.3	26.8	38	4	0	11.9
U2 - Beta(0.5, 0.5)	18.9	28.7	36.3	4.2	0	11.9
U2 - Beta(0.7, 0.3)	19.5	28.5	35.8	4	0	12.2
U2 - Beta(1, 1)	21.5	27.2	35.2	4.3	0	11.8
U3 - Beta(0.3, 0.7)	21.8	27	35.1	4.1	0	12
U3 - Beta(0.5, 0.5)	23.8	27.9	32.3	3.9	0	12.1
U3 - Beta(0.7, 0.3)	23.2	28.8	31.9	4.2	0	11.9
U3 - Beta(1, 1)	24.7	28.2	32.1	3.3	0	11.7
Mean administrations						
U2 - Beta(0.3, 0.7)	8.2	6.9	9.1	3.2	0.8	-
U2 - Beta(0.5, 0.5)	8.2	7	9.1	3.1	0.8	-
U2 - Beta(0.7, 0.3)	8.2	7.1	9	3.2	0.7	-
U2 - Beta(1, 1)	8.2	7.1	9	3.1	0.8	-
U3 - Beta(0.3, 0.7)	8.1	7	9.1	3.1	0.8	-
U3 - Beta(0.5, 0.5)	8.2	7	9.1	3.1	0.7	-
U3 - Beta(0.7, 0.3)	8.3	7.1	8.9	3.1	0.7	-
U3 - Beta(1, 1)	8.2	7.2	8.9	3.1	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.2	3.4	2.6	1	0.2	-
U2 - Beta(0.5, 0.5)	2.2	3.4	2.6	1	0.2	-
U2 - Beta(0.7, 0.3)	2.2	3.4	2.6	1	0.2	-
U2 - Beta(1, 1)	2.2	3.4	2.6	1	0.2	-
U3 - Beta(0.3, 0.7)	2.2	3.4	2.6	1	0.2	-
U3 - Beta(0.5, 0.5)	2.2	3.4	2.6	1	0.2	-
U3 - Beta(0.7, 0.3)	2.3	3.4	2.6	1	0.2	-
U3 - Beta(1, 1)	2.3	3.4	2.6	1	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1030.6
U2 - Beta(0.5, 0.5)						1029.9
U2 - Beta(0.7, 0.3)						1028.6
U2 - Beta(1, 1)						1029.5
U3 - Beta(0.3, 0.7)						1030.8
U3 - Beta(0.5, 0.5)						1029.4
U3 - Beta(0.7, 0.3)						1027.8
U3 - Beta(1, 1)						1029.8

Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	18.4	27.6	1.3	0	0	52.7
U2 - Beta(0.5, 0.5)	18.5	28	1.2	0	0	52.3
U2 - Beta(0.7, 0.3)	18.5	27.3	1.7	0	0	52.5
U2 - Beta(1, 1)	17.8	28.4	1.5	0	0	52.3
U3 - Beta(0.3, 0.7)	18.7	27.9	1.1	0	0	52.3
U3 - Beta(0.5, 0.5)	18.8	27.1	1.3	0	0	52.8
U3 - Beta(0.7, 0.3)	19	27.5	1.3	0	0	52.2
U3 - Beta(1, 1)	20	26.9	1.1	0	0	52
Mean administrations						
U2 - Beta(0.3, 0.7)	10.2	7.5	2.9	0.5	0	-
U2 - Beta(0.5, 0.5)	10.3	7.5	2.9	0.5	0	-
U2 - Beta(0.7, 0.3)	10.3	7.4	2.9	0.4	0	-
U2 - Beta(1, 1)	10.3	7.4	2.9	0.5	0	-
U3 - Beta(0.3, 0.7)	10.2	7.5	2.9	0.5	0	-
U3 - Beta(0.5, 0.5)	10.2	7.5	2.9	0.5	0	-
U3 - Beta(0.7, 0.3)	10.3	7.4	2.9	0.4	0	-
U3 - Beta(1, 1)	10.3	7.4	2.9	0.5	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.9	3.2	0.9	0.2	0	-
U2 - Beta(0.5, 0.5)	2.9	3.2	0.9	0.2	0	-
U2 - Beta(0.7, 0.3)	2.9	3.2	0.9	0.2	0	-
U2 - Beta(1, 1)	2.9	3.2	0.9	0.2	0	-
U3 - Beta(0.3, 0.7)	2.9	3.2	0.9	0.2	0	-
U3 - Beta(0.5, 0.5)	2.9	3.2	0.9	0.2	0	-
U3 - Beta(0.7, 0.3)	2.9	3.2	0.9	0.2	0	-
U3 - Beta(1, 1)	2.9	3.2	0.9	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						762.4
U2 - Beta(0.5, 0.5)						761.6
U2 - Beta(0.7, 0.3)						761.1
U2 - Beta(1, 1)						762.2
U3 - Beta(0.3, 0.7)						762
U3 - Beta(0.5, 0.5)						762.2
U3 - Beta(0.7, 0.3)						760.8
U3 - Beta(1, 1)						761

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	74.6	20.6	1.4	0.1	0	3.3
U2 - Beta(0.5, 0.5)	75.3	20	1.4	0	0	3.3
U2 - Beta(0.7, 0.3)	73.8	21.6	1.4	0	0	3.2
U2 - Beta(1, 1)	75.4	20	1.4	0	0	3.2
U3 - Beta(0.3, 0.7)	78.3	17.8	0.6	0	0	3.3
U3 - Beta(0.5, 0.5)	76.6	18.5	1.4	0	0	3.5
U3 - Beta(0.7, 0.3)	75.5	19.8	1.4	0	0	3.3
U3 - Beta(1, 1)	76.9	19.1	0.8	0	0	3.2
Mean administrations						
U2 - Beta(0.3, 0.7)	19.4	7	2.6	0.3	0	-
U2 - Beta(0.5, 0.5)	19.5	6.9	2.6	0.3	0	-
U2 - Beta(0.7, 0.3)	19.5	7	2.6	0.3	0	-
U2 - Beta(1, 1)	19.4	7	2.6	0.3	0	-
U3 - Beta(0.3, 0.7)	19.5	6.9	2.6	0.3	0	-
U3 - Beta(0.5, 0.5)	19.5	6.9	2.6	0.3	0	-
U3 - Beta(0.7, 0.3)	19.5	6.9	2.6	0.3	0	-
U3 - Beta(1, 1)	19.5	7	2.6	0.3	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.1	2.4	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	5.1	2.3	0.7	0.1	0	-
U2 - Beta(0.7, 0.3)	5.1	2.4	0.7	0.1	0	-
U2 - Beta(1, 1)	5.1	2.3	0.7	0.1	0	-
U3 - Beta(0.3, 0.7)	5.1	2.4	0.7	0.1	0	-
U3 - Beta(0.5, 0.5)	5.1	2.3	0.7	0.1	0	-
U3 - Beta(0.7, 0.3)	5.1	2.3	0.7	0.1	0	-
U3 - Beta(1, 1)	5.1	2.4	0.7	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1166
U2 - Beta(0.5, 0.5)						1164.8
U2 - Beta(0.7, 0.3)						1165.6
U2 - Beta(1, 1)						1165.4
U3 - Beta(0.3, 0.7)						1166.1
U3 - Beta(0.5, 0.5)						1164.9
U3 - Beta(0.7, 0.3)						1164.9
U3 - Beta(1, 1)						1165.4

Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	17.8	17.2	19.8	36.4	6.9	1.9
U2 - Beta(0.5, 0.5)	19.2	17.4	18.6	36	6.9	1.9
U2 - Beta(0.7, 0.3)	19.8	19	16.7	34.5	8	2
U2 - Beta(1, 1)	20.5	17.6	17.2	35.4	7.3	2
U3 - Beta(0.3, 0.7)	23.5	19.1	16.5	33.3	5.7	1.9
U3 - Beta(0.5, 0.5)	24.7	20.5	15.3	31.7	5.9	1.9
U3 - Beta(0.7, 0.3)	26.3	20.6	13.5	31.2	6.5	1.9
U3 - Beta(1, 1)	27.8	21.8	13	30.1	5.3	2
Mean administrations						
U2 - Beta(0.3, 0.7)	5.2	4.8	6.3	8.3	5	-
U2 - Beta(0.5, 0.5)	5.2	4.9	6.5	8	5	-
U2 - Beta(0.7, 0.3)	5.3	5.1	6.4	8	4.9	-
U2 - Beta(1, 1)	5.4	4.9	6.4	8.1	5	-
U3 - Beta(0.3, 0.7)	5.2	4.8	6.4	8.2	5.1	-
U3 - Beta(0.5, 0.5)	5.3	5	6.4	8	5	-
U3 - Beta(0.7, 0.3)	5.3	5	6.4	8.2	4.8	-
U3 - Beta(1, 1)	5.3	5	6.4	8	4.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1	2.8	2.1	2.9	1.3	-
U2 - Beta(0.5, 0.5)	1	2.9	2.1	2.8	1.3	-
U2 - Beta(0.7, 0.3)	1	3	2.1	2.8	1.3	-
U2 - Beta(1, 1)	1.1	2.9	2	2.8	1.3	-
U3 - Beta(0.3, 0.7)	1	2.9	2.1	2.9	1.3	-
U3 - Beta(0.5, 0.5)	1	2.9	2.1	2.8	1.3	-
U3 - Beta(0.7, 0.3)	1	2.9	2.1	2.8	1.3	-
U3 - Beta(1, 1)	1	2.9	2.1	2.8	1.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1081.5
U2 - Beta(0.5, 0.5)						1081.9
U2 - Beta(0.7, 0.3)						1079.6
U2 - Beta(1, 1)						1081
U3 - Beta(0.3, 0.7)						1082.2
U3 - Beta(0.5, 0.5)						1080.4
U3 - Beta(0.7, 0.3)						1079.4
U3 - Beta(1, 1)						1080.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.7	34.3	30.5	25.5	4.8	0.2
U2 - Beta(0.5, 0.5)	5.1	37.3	28.8	23.2	5.4	0.2
U2 - Beta(0.7, 0.3)	4.9	38	28.5	22.9	5.5	0.2
U2 - Beta(1, 1)	5.1	37.8	29.1	22.8	5	0.2
U3 - Beta(0.3, 0.7)	6.7	42.4	27.9	19.9	2.9	0.2
U3 - Beta(0.5, 0.5)	7.8	44.3	25.5	18.2	4	0.2
U3 - Beta(0.7, 0.3)	7.1	45.5	25.3	17.8	4.1	0.2
U3 - Beta(1, 1)	7.6	46.3	25.2	17.4	3.3	0.2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.1	5.7	6.6	8	5.6	-
U2 - Beta(0.5, 0.5)	4.1	5.9	6.5	8	5.4	-
U2 - Beta(0.7, 0.3)	4.2	6	6.5	7.9	5.4	-
U2 - Beta(1, 1)	4.1	5.9	6.4	8	5.5	-
U3 - Beta(0.3, 0.7)	4.1	5.8	6.6	8	5.5	-
U3 - Beta(0.5, 0.5)	4.2	5.9	6.5	7.9	5.5	-
U3 - Beta(0.7, 0.3)	4.2	5.9	6.6	7.9	5.4	-
U3 - Beta(1, 1)	4.1	5.8	6.5	8	5.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.5	2.9	1.7	2	1.4	-
U2 - Beta(0.5, 0.5)	0.5	2.9	1.7	2	1.4	-
U2 - Beta(0.7, 0.3)	0.5	2.9	1.7	2	1.4	-
U2 - Beta(1, 1)	0.5	2.9	1.6	2	1.4	-
U3 - Beta(0.3, 0.7)	0.5	2.9	1.7	2	1.4	-
U3 - Beta(0.5, 0.5)	0.5	2.9	1.7	2	1.4	-
U3 - Beta(0.7, 0.3)	0.5	2.9	1.7	2	1.4	-
U3 - Beta(1, 1)	0.5	2.9	1.7	2	1.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1184.9
U2 - Beta(0.5, 0.5)						1183.5
U2 - Beta(0.7, 0.3)						1182.1
U2 - Beta(1, 1)						1184.1
U3 - Beta(0.3, 0.7)						1185
U3 - Beta(0.5, 0.5)						1182.9
U3 - Beta(0.7, 0.3)						1183
U3 - Beta(1, 1)						1185.1
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.9	13.9	36.5	29.2	11.8	0.7
U2 - Beta(0.5, 0.5)	9.2	15.3	35.1	30	9.7	0.7
U2 - Beta(0.7, 0.3)	10.4	15.7	33.5	30.5	9.2	0.7
U2 - Beta(1, 1)	10.8	15.9	34.5	29.7	8.4	0.7
U3 - Beta(0.3, 0.7)	13.5	20.1	33.1	25.8	6.8	0.7
U3 - Beta(0.5, 0.5)	13.3	20	34.1	25.9	6	0.7
U3 - Beta(0.7, 0.3)	16.1	21	33.4	24.9	3.9	0.7
U3 - Beta(1, 1)	18	20.3	32.5	24.7	3.8	0.7
Mean administrations						
U2 - Beta(0.3, 0.7)	3.8	4.3	7.9	7.6	6.4	-
U2 - Beta(0.5, 0.5)	3.9	4.4	7.8	7.7	6.1	-
U2 - Beta(0.7, 0.3)	4	4.5	7.8	7.7	6	-
U2 - Beta(1, 1)	3.9	4.4	7.9	7.7	6.1	-
U3 - Beta(0.3, 0.7)	3.8	4.3	7.7	7.8	6.3	-
U3 - Beta(0.5, 0.5)	3.9	4.4	7.8	7.7	6.1	-
U3 - Beta(0.7, 0.3)	3.9	4.4	7.9	7.6	6.1	-
U3 - Beta(1, 1)	3.9	4.4	7.8	7.7	6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.8	2.5	2.8	2.2	-
U2 - Beta(0.5, 0.5)	0.4	2.9	2.5	2.9	2	-
U2 - Beta(0.7, 0.3)	0.4	2.9	2.5	2.9	2	-
U2 - Beta(1, 1)	0.4	2.9	2.5	2.8	2.1	-
U3 - Beta(0.3, 0.7)	0.4	2.8	2.5	2.9	2.1	-
U3 - Beta(0.5, 0.5)	0.4	2.9	2.5	2.9	2.1	-
U3 - Beta(0.7, 0.3)	0.4	2.9	2.5	2.8	2.1	-
U3 - Beta(1, 1)	0.4	2.9	2.5	2.9	2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1057.3
U2 - Beta(0.5, 0.5)						1056.9
U2 - Beta(0.7, 0.3)						1055.8
U2 - Beta(1, 1)						1056.7
U3 - Beta(0.3, 0.7)						1058
U3 - Beta(0.5, 0.5)						1056.8
U3 - Beta(0.7, 0.3)						1056.3
U3 - Beta(1, 1)						1054.9

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.2	7.6	15.9	43.1	25.1	0.1
U2 - Beta(0.5, 0.5)	9	8.2	15	42.9	24.9	0
U2 - Beta(0.7, 0.3)	8.6	9.2	15.2	43.9	23	0.1
U2 - Beta(1, 1)	9.8	8.5	14.2	42.9	24.5	0.1
U3 - Beta(0.3, 0.7)	10.7	9.8	15.5	41.4	22.5	0.1
U3 - Beta(0.5, 0.5)	13.2	10	15.1	39.3	22.3	0.1
U3 - Beta(0.7, 0.3)	13.3	11.2	14.6	41.9	18.9	0.1
U3 - Beta(1, 1)	14.7	10.7	14.4	39.6	20.5	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.4	3.6	5.5	8.1	9.4	-
U2 - Beta(0.5, 0.5)	3.5	3.6	5.4	8.2	9.3	-
U2 - Beta(0.7, 0.3)	3.5	3.7	5.4	8.3	9.1	-
U2 - Beta(1, 1)	3.5	3.7	5.3	8.1	9.4	-
U3 - Beta(0.3, 0.7)	3.4	3.6	5.5	8	9.5	-
U3 - Beta(0.5, 0.5)	3.5	3.7	5.4	8.1	9.3	-
U3 - Beta(0.7, 0.3)	3.5	3.7	5.4	8.2	9.2	-
U3 - Beta(1, 1)	3.5	3.7	5.4	8.2	9.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.2	2.6	1.7	3.1	2.3	-
U2 - Beta(0.5, 0.5)	0.2	2.6	1.7	3	2.3	-
U2 - Beta(0.7, 0.3)	0.3	2.7	1.7	3.1	2.3	-
U2 - Beta(1, 1)	0.3	2.7	1.7	3	2.3	-
U3 - Beta(0.3, 0.7)	0.2	2.6	1.7	3	2.3	-
U3 - Beta(0.5, 0.5)	0.2	2.6	1.7	3	2.3	-
U3 - Beta(0.7, 0.3)	0.2	2.7	1.7	3.1	2.3	-
U3 - Beta(1, 1)	0.2	2.7	1.7	3.1	2.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1105.4
U2 - Beta(0.5, 0.5)						1105.2
U2 - Beta(0.7, 0.3)						1103.6
U2 - Beta(1, 1)						1105.1
U3 - Beta(0.3, 0.7)						1106.1
U3 - Beta(0.5, 0.5)						1104.3
U3 - Beta(0.7, 0.3)						1103.9
U3 - Beta(1, 1)						1103.3
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.9	16	46.4	14.9	2.9	5.9
U2 - Beta(0.5, 0.5)	14.6	15.1	47.6	14.2	2.8	5.7
U2 - Beta(0.7, 0.3)	15.2	16.1	47.9	12.3	2.6	5.9
U2 - Beta(1, 1)	17.7	15.3	46.3	12.5	2.4	5.8
U3 - Beta(0.3, 0.7)	17.9	17.3	44.8	11.9	2.2	5.9
U3 - Beta(0.5, 0.5)	19.4	16.8	44.3	11.7	1.9	5.9
U3 - Beta(0.7, 0.3)	21.9	16.7	42	11.8	1.7	5.9
U3 - Beta(1, 1)	24.8	17	39.9	11.3	1.5	5.5
Mean administrations						
U2 - Beta(0.3, 0.7)	5.3	6.2	10.3	4.9	2.4	-
U2 - Beta(0.5, 0.5)	5.5	6.2	10.2	4.9	2.3	-
U2 - Beta(0.7, 0.3)	5.4	6.3	10.1	4.9	2.3	-
U2 - Beta(1, 1)	5.5	6.3	10	4.9	2.4	-
U3 - Beta(0.3, 0.7)	5.2	6.2	10.2	4.9	2.4	-
U3 - Beta(0.5, 0.5)	5.4	6.2	10.2	4.8	2.3	-
U3 - Beta(0.7, 0.3)	5.4	6.4	10	4.9	2.3	-
U3 - Beta(1, 1)	5.4	6.3	10.1	4.9	2.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.1	3.4	3.3	1.7	0.8	-
U2 - Beta(0.5, 0.5)	1.1	3.4	3.2	1.7	0.7	-
U2 - Beta(0.7, 0.3)	1.1	3.5	3.2	1.7	0.7	-
U2 - Beta(1, 1)	1.1	3.5	3.2	1.7	0.8	-
U3 - Beta(0.3, 0.7)	1	3.4	3.2	1.7	0.8	-
U3 - Beta(0.5, 0.5)	1.1	3.4	3.3	1.7	0.8	-
U3 - Beta(0.7, 0.3)	1.1	3.6	3.2	1.7	0.7	-
U3 - Beta(1, 1)	1.1	3.5	3.2	1.7	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1031.3
U2 - Beta(0.5, 0.5)						1029.4
U2 - Beta(0.7, 0.3)						1030
U2 - Beta(1, 1)						1030.7
U3 - Beta(0.3, 0.7)						1032.3
U3 - Beta(0.5, 0.5)						1029.6
U3 - Beta(0.7, 0.3)						1029.4
U3 - Beta(1, 1)						1029.7

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10	46.7	32.1	9.5	1.5	0.2
U2 - Beta(0.5, 0.5)	11.7	44.6	32.1	9.8	1.6	0.2
U2 - Beta(0.7, 0.3)	12.4	45.2	31.5	9.1	1.6	0.2
U2 - Beta(1, 1)	12.2	44.7	31.9	9.8	1.2	0.2
U3 - Beta(0.3, 0.7)	12.9	52.3	27	6.8	0.8	0.2
U3 - Beta(0.5, 0.5)	14.8	50.2	26	8	0.8	0.2
U3 - Beta(0.7, 0.3)	15.8	49.8	24.9	8.4	0.9	0.2
U3 - Beta(1, 1)	15.8	51.2	24.9	7.2	0.7	0.2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.5	9	9.3	4.9	2.3	-
U2 - Beta(0.5, 0.5)	4.6	9.1	9.3	4.8	2.2	-
U2 - Beta(0.7, 0.3)	4.6	9.3	9.2	4.7	2.2	-
U2 - Beta(1, 1)	4.5	9.1	9.3	4.8	2.3	-
U3 - Beta(0.3, 0.7)	4.5	9	9.4	4.8	2.3	-
U3 - Beta(0.5, 0.5)	4.5	9.1	9.3	4.8	2.3	-
U3 - Beta(0.7, 0.3)	4.6	9.1	9.2	4.8	2.2	-
U3 - Beta(1, 1)	4.6	9.1	9.3	4.8	2.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	3.8	2.3	1.4	0.6	-
U2 - Beta(0.5, 0.5)	0.7	3.9	2.3	1.4	0.7	-
U2 - Beta(0.7, 0.3)	0.7	3.9	2.3	1.4	0.6	-
U2 - Beta(1, 1)	0.6	3.9	2.3	1.4	0.6	-
U3 - Beta(0.3, 0.7)	0.6	3.8	2.4	1.4	0.6	-
U3 - Beta(0.5, 0.5)	0.6	3.9	2.3	1.4	0.7	-
U3 - Beta(0.7, 0.3)	0.7	3.9	2.3	1.4	0.6	-
U3 - Beta(1, 1)	0.6	3.9	2.3	1.4	0.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1164.3
U2 - Beta(0.5, 0.5)						1162.7
U2 - Beta(0.7, 0.3)						1161.8
U2 - Beta(1, 1)						1163.1
U3 - Beta(0.3, 0.7)						1163.6
U3 - Beta(0.5, 0.5)						1163.2
U3 - Beta(0.7, 0.3)						1162.7
U3 - Beta(1, 1)						1163.9

Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	19.1	37.8	10.1	2.5	0.7	29.8
U2 - Beta(0.5, 0.5)	17.1	38.3	10.3	2.8	0.9	30.6
U2 - Beta(0.7, 0.3)	19.3	37.8	10	2.3	0.9	29.7
U2 - Beta(1, 1)	20.9	36.7	9.9	2.4	0.8	29.3
U3 - Beta(0.3, 0.7)	20.8	36.7	9.3	2.4	0.8	30
U3 - Beta(0.5, 0.5)	22.9	36.5	8.2	2	0.8	29.6
U3 - Beta(0.7, 0.3)	23.3	36.2	8	2.3	0.6	29.6
U3 - Beta(1, 1)	25.7	34.5	7	2.5	0.6	29.7
Mean administrations						
U2 - Beta(0.3, 0.7)	8.9	8.4	5	1.9	0.7	-
U2 - Beta(0.5, 0.5)	9	8.3	5	1.9	0.7	-
U2 - Beta(0.7, 0.3)	9	8.5	4.8	1.9	0.7	-
U2 - Beta(1, 1)	9	8.4	4.9	1.9	0.7	-
U3 - Beta(0.3, 0.7)	8.9	8.4	5	1.9	0.7	-
U3 - Beta(0.5, 0.5)	8.9	8.5	4.9	1.8	0.7	-
U3 - Beta(0.7, 0.3)	9	8.4	4.9	1.9	0.7	-
U3 - Beta(1, 1)	9.1	8.4	4.9	1.9	0.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	4	1.6	0.8	0.2	-
U2 - Beta(0.5, 0.5)	2.5	3.9	1.6	0.8	0.2	-
U2 - Beta(0.7, 0.3)	2.5	4	1.6	0.7	0.2	-
U2 - Beta(1, 1)	2.5	4	1.6	0.7	0.2	-
U3 - Beta(0.3, 0.7)	2.4	4	1.6	0.7	0.2	-
U3 - Beta(0.5, 0.5)	2.4	4	1.6	0.7	0.2	-
U3 - Beta(0.7, 0.3)	2.5	4	1.6	0.8	0.2	-
U3 - Beta(1, 1)	2.5	4	1.6	0.7	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						875.7
U2 - Beta(0.5, 0.5)						874.3
U2 - Beta(0.7, 0.3)						874.7
U2 - Beta(1, 1)						874.5
U3 - Beta(0.3, 0.7)						874.6
U3 - Beta(0.5, 0.5)						874.6
U3 - Beta(0.7, 0.3)						874
U3 - Beta(1, 1)						875.1

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	22.9	46.3	10.2	1.1	0.1	19.4
U2 - Beta(0.5, 0.5)	24	44.5	10.1	1.2	0.2	20
U2 - Beta(0.7, 0.3)	23.4	45.5	10.2	1	0.1	19.8
U2 - Beta(1, 1)	24.6	44.8	10.3	0.9	0.1	19.3
U3 - Beta(0.3, 0.7)	27.6	42.3	8.4	1	0.1	20.6
U3 - Beta(0.5, 0.5)	28.1	42.5	8.2	0.9	0.1	20.2
U3 - Beta(0.7, 0.3)	28.6	43.1	7.5	0.7	0	20.1
U3 - Beta(1, 1)	29.9	43	6.4	0.8	0.1	19.8
Mean administrations						
U2 - Beta(0.3, 0.7)	8.3	10.7	5.6	1.5	0.5	-
U2 - Beta(0.5, 0.5)	8.3	10.7	5.5	1.5	0.4	-
U2 - Beta(0.7, 0.3)	8.4	10.9	5.3	1.5	0.4	-
U2 - Beta(1, 1)	8.4	10.9	5.4	1.4	0.4	-
U3 - Beta(0.3, 0.7)	8.2	10.7	5.7	1.5	0.5	-
U3 - Beta(0.5, 0.5)	8.3	10.8	5.6	1.4	0.5	-
U3 - Beta(0.7, 0.3)	8.4	10.8	5.4	1.4	0.5	-
U3 - Beta(1, 1)	8.5	10.7	5.5	1.5	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	5	1.7	0.6	0.2	-
U2 - Beta(0.5, 0.5)	2.1	5	1.7	0.7	0.2	-
U2 - Beta(0.7, 0.3)	2.1	5.1	1.7	0.7	0.2	-
U2 - Beta(1, 1)	2.1	5.1	1.7	0.6	0.2	-
U3 - Beta(0.3, 0.7)	2.1	5	1.8	0.6	0.2	-
U3 - Beta(0.5, 0.5)	2.1	5	1.8	0.6	0.2	-
U3 - Beta(0.7, 0.3)	2.1	5	1.7	0.6	0.2	-
U3 - Beta(1, 1)	2.2	5	1.7	0.6	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						927.1
U2 - Beta(0.5, 0.5)						924.9
U2 - Beta(0.7, 0.3)						925.3
U2 - Beta(1, 1)						925.4
U3 - Beta(0.3, 0.7)						925.6
U3 - Beta(0.5, 0.5)						927.1
U3 - Beta(0.7, 0.3)						924.6
U3 - Beta(1, 1)						925.2

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.3	4.3	1.1	0	0	82.3
U2 - Beta(0.5, 0.5)	14.8	4.2	0.9	0	0	80.1
U2 - Beta(0.7, 0.3)	14.2	5.1	0.3	0	0	80.4
U2 - Beta(1, 1)	14.6	4.9	0.4	0	0	80.1
U3 - Beta(0.3, 0.7)	11.8	3.8	0.7	0	0	83.7
U3 - Beta(0.5, 0.5)	14.4	4.5	0.7	0	0	80.4
U3 - Beta(0.7, 0.3)	14.3	4.7	0.7	0.1	0	80.2
U3 - Beta(1, 1)	14.6	4.1	0.9	0	0	80.4
Mean administrations						
U2 - Beta(0.3, 0.7)	11.5	4.7	1.3	0.5	0.1	-
U2 - Beta(0.5, 0.5)	11.5	4.8	1.4	0.6	0.1	-
U2 - Beta(0.7, 0.3)	11.5	4.9	1.4	0.6	0.1	-
U2 - Beta(1, 1)	11.4	4.9	1.5	0.6	0.1	-
U3 - Beta(0.3, 0.7)	11.6	4.6	1.3	0.4	0.1	-
U3 - Beta(0.5, 0.5)	11.6	4.8	1.4	0.6	0.1	-
U3 - Beta(0.7, 0.3)	11.5	4.8	1.4	0.6	0.1	-
U3 - Beta(1, 1)	11.6	4.8	1.4	0.6	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	2.5	0.5	0.2	0	-
U2 - Beta(0.5, 0.5)	3.1	2.6	0.5	0.3	0	-
U2 - Beta(0.7, 0.3)	3.1	2.6	0.5	0.3	0	-
U2 - Beta(1, 1)	3.1	2.6	0.5	0.3	0	-
U3 - Beta(0.3, 0.7)	3.2	2.5	0.5	0.2	0	-
U3 - Beta(0.5, 0.5)	3.2	2.6	0.5	0.3	0	-
U3 - Beta(0.7, 0.3)	3.2	2.6	0.5	0.3	0	-
U3 - Beta(1, 1)	3.2	2.6	0.5	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						633.4
U2 - Beta(0.5, 0.5)						646.9
U2 - Beta(0.7, 0.3)						648
U2 - Beta(1, 1)						647.4
U3 - Beta(0.3, 0.7)						630.6
U3 - Beta(0.5, 0.5)						645.1
U3 - Beta(0.7, 0.3)						648.3
U3 - Beta(1, 1)						647.1
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	33.3	35.4	26.4	1.8	0	3.1
U2 - Beta(0.5, 0.5)	28.9	23.8	36.8	7.2	0.5	2.8
U2 - Beta(0.7, 0.3)	29.8	25.4	32.8	8.3	0.9	2.8
U2 - Beta(1, 1)	27.5	25.1	37	7.4	0.4	2.6
U3 - Beta(0.3, 0.7)	42.8	35.4	17.5	1.2	0	3.1
U3 - Beta(0.5, 0.5)	33.7	24	34	5.3	0.3	2.7
U3 - Beta(0.7, 0.3)	31.7	25	30.8	8.9	0.7	2.9
U3 - Beta(1, 1)	32.8	24.5	34.8	5	0.3	2.6
Mean administrations						
U2 - Beta(0.3, 0.7)	13.7	10.3	5	0.4	0	-
U2 - Beta(0.5, 0.5)	10.6	8.3	8.3	2.2	0.2	-
U2 - Beta(0.7, 0.3)	10.6	7.7	7.1	3.5	0.8	-
U2 - Beta(1, 1)	10.4	8.5	8.2	2.3	0.2	-
U3 - Beta(0.3, 0.7)	15.2	10.3	3.7	0.3	0	-
U3 - Beta(0.5, 0.5)	11.1	8.5	7.9	2	0.2	-
U3 - Beta(0.7, 0.3)	10.7	8.1	7.2	3.2	0.4	-
U3 - Beta(1, 1)	11.1	8.6	7.8	1.9	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.1	4.4	1.6	0.1	0	-
U2 - Beta(0.5, 0.5)	2.7	3.7	2.4	0.7	0.1	-
U2 - Beta(0.7, 0.3)	2.8	3.5	1.9	1	0.3	-
U2 - Beta(1, 1)	2.6	3.8	2.4	0.7	0.1	-
U3 - Beta(0.3, 0.7)	4.7	4.4	1.2	0.1	0	-
U3 - Beta(0.5, 0.5)	2.9	3.8	2.3	0.6	0.1	-
U3 - Beta(0.7, 0.3)	2.8	3.6	2	1	0.1	-
U3 - Beta(1, 1)	3	3.8	2.2	0.6	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1061.6
U2 - Beta(0.5, 0.5)						1102
U2 - Beta(0.7, 0.3)						1111.4
U2 - Beta(1, 1)						1103.6
U3 - Beta(0.3, 0.7)						1051.1
U3 - Beta(0.5, 0.5)						1096.7
U3 - Beta(0.7, 0.3)						1104.9
U3 - Beta(1, 1)						1095.9

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.3	67.4	12	1	0	11.3
U2 - Beta(0.5, 0.5)	9.5	50.8	23.2	5.5	0.4	10.6
U2 - Beta(0.7, 0.3)	9.1	51.3	21.6	6.7	1	10.3
U2 - Beta(1, 1)	8.6	52.1	22.8	5.3	0.5	10.7
U3 - Beta(0.3, 0.7)	9.1	69.8	9.2	0.6	0	11.3
U3 - Beta(0.5, 0.5)	9.9	55.2	20.6	3.3	0.4	10.6
U3 - Beta(0.7, 0.3)	10.7	51.9	20.8	5.7	0.7	10.2
U3 - Beta(1, 1)	9.8	55.1	20.6	3.6	0.4	10.5
Mean administrations						
U2 - Beta(0.3, 0.7)	8.4	16.9	2.9	0.3	0	-
U2 - Beta(0.5, 0.5)	7.5	12.4	6.1	2.3	0.4	-
U2 - Beta(0.7, 0.3)	7.4	10.8	6.3	3.4	0.9	-
U2 - Beta(1, 1)	7.3	12.5	6.1	2.3	0.4	-
U3 - Beta(0.3, 0.7)	8.9	17.1	2.2	0.2	0	-
U3 - Beta(0.5, 0.5)	7.7	13.1	5.6	2	0.3	-
U3 - Beta(0.7, 0.3)	7.5	11.5	6	3	0.6	-
U3 - Beta(1, 1)	7.7	13.2	5.5	1.9	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	6.6	0.9	0.1	0	-
U2 - Beta(0.5, 0.5)	1.5	5.1	1.8	0.8	0.1	-
U2 - Beta(0.7, 0.3)	1.5	4.6	1.8	1.1	0.3	-
U2 - Beta(1, 1)	1.4	5.1	1.8	0.8	0.1	-
U3 - Beta(0.3, 0.7)	2.2	6.6	0.7	0.1	0	-
U3 - Beta(0.5, 0.5)	1.6	5.3	1.7	0.7	0.1	-
U3 - Beta(0.7, 0.3)	1.5	4.8	1.8	1	0.2	-
U3 - Beta(1, 1)	1.6	5.4	1.6	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1039.2
U2 - Beta(0.5, 0.5)						1055.9
U2 - Beta(0.7, 0.3)						1063.3
U2 - Beta(1, 1)						1056.8
U3 - Beta(0.3, 0.7)						1035.1
U3 - Beta(0.5, 0.5)						1052.7
U3 - Beta(0.7, 0.3)						1060.1
U3 - Beta(1, 1)						1052.2
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.3	8.6	37.1	46.1	1.8	2.1
U2 - Beta(0.5, 0.5)	6.1	7.6	23.6	52.6	8.3	1.8
U2 - Beta(0.7, 0.3)	7	8.7	22.9	49.5	10.1	1.8
U2 - Beta(1, 1)	7.4	7.8	21.6	53.8	7.7	1.7
U3 - Beta(0.3, 0.7)	7.2	10.1	36.3	43.7	0.6	2.1
U3 - Beta(0.5, 0.5)	9.4	8.6	21.2	53.4	5.5	1.9
U3 - Beta(0.7, 0.3)	8.8	9	20.6	50.4	9.3	1.9
U3 - Beta(1, 1)	10.6	8.9	19.7	53.9	5	1.9
Mean administrations						
U2 - Beta(0.3, 0.7)	4.5	5.9	10.9	8.1	0.4	-
U2 - Beta(0.5, 0.5)	4.3	5	7.7	10.5	2.4	-
U2 - Beta(0.7, 0.3)	4.4	5.1	7.3	9.5	3.6	-
U2 - Beta(1, 1)	4.3	5	7.6	10.6	2.3	-
U3 - Beta(0.3, 0.7)	4.9	6.4	10.8	7.5	0.2	-
U3 - Beta(0.5, 0.5)	4.5	5.1	7.7	10.5	2	-
U3 - Beta(0.7, 0.3)	4.7	5.3	7.1	9.6	3.1	-
U3 - Beta(1, 1)	4.7	5.3	7.4	10.5	1.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.4	3.9	2.7	0.1	-
U2 - Beta(0.5, 0.5)	0.5	3.2	2.3	3.4	0.7	-
U2 - Beta(0.7, 0.3)	0.6	3.2	2.1	3.1	1	-
U2 - Beta(1, 1)	0.5	3.2	2.2	3.4	0.7	-
U3 - Beta(0.3, 0.7)	0.9	3.5	3.9	2.5	0.1	-
U3 - Beta(0.5, 0.5)	0.6	3.2	2.3	3.4	0.6	-
U3 - Beta(0.7, 0.3)	0.7	3.3	2.1	3.1	0.9	-
U3 - Beta(1, 1)	0.7	3.3	2.2	3.4	0.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1048.4
U2 - Beta(0.5, 0.5)						1089
U2 - Beta(0.7, 0.3)						1093
U2 - Beta(1, 1)						1087
U3 - Beta(0.3, 0.7)						1039.1
U3 - Beta(0.5, 0.5)						1083.8
U3 - Beta(0.7, 0.3)						1085.2
U3 - Beta(1, 1)						1081.3

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	1.5	49.2	29	15.1	3.1	2.1
U2 - Beta(0.5, 0.5)	5.3	35	25.9	23.1	8.8	1.9
U2 - Beta(0.7, 0.3)	4.9	39.5	26	21.1	6.6	1.9
U2 - Beta(1, 1)	5.5	36.7	25.7	22.2	8	1.9
U3 - Beta(0.3, 0.7)	2.9	54.2	26.5	12.5	1.8	2.1
U3 - Beta(0.5, 0.5)	7.6	41.6	24.7	18.2	6	1.9
U3 - Beta(0.7, 0.3)	6.5	42.8	25	18.9	5	1.8
U3 - Beta(1, 1)	8.6	43.8	23.7	17.8	4.3	1.8
Mean administrations						
U2 - Beta(0.3, 0.7)	4.5	14.7	7	2.9	0.6	-
U2 - Beta(0.5, 0.5)	4.1	8.1	7.3	6.9	3.2	-
U2 - Beta(0.7, 0.3)	4.1	7.2	7.3	7.2	3.9	-
U2 - Beta(1, 1)	4.1	8.2	7.4	6.8	3.3	-
U3 - Beta(0.3, 0.7)	4.8	15.4	6.5	2.6	0.4	-
U3 - Beta(0.5, 0.5)	4.2	8.8	7.5	6.5	2.7	-
U3 - Beta(0.7, 0.3)	4.2	8	7.4	6.9	3.2	-
U3 - Beta(1, 1)	4.2	9.2	7.4	6.4	2.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	6.5	2.6	1.1	0.2	-
U2 - Beta(0.5, 0.5)	0.4	4.1	2.4	2.6	1.2	-
U2 - Beta(0.7, 0.3)	0.4	3.9	2.3	2.8	1.3	-
U2 - Beta(1, 1)	0.4	4.1	2.4	2.6	1.2	-
U3 - Beta(0.3, 0.7)	0.8	6.7	2.4	1	0.1	-
U3 - Beta(0.5, 0.5)	0.4	4.4	2.5	2.5	1	-
U3 - Beta(0.7, 0.3)	0.4	4.2	2.4	2.5	1.2	-
U3 - Beta(1, 1)	0.4	4.6	2.5	2.4	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1028.6
U2 - Beta(0.5, 0.5)						1044.1
U2 - Beta(0.7, 0.3)						1046.8
U2 - Beta(1, 1)						1044.5
U3 - Beta(0.3, 0.7)						1026.5
U3 - Beta(0.5, 0.5)						1041.3
U3 - Beta(0.7, 0.3)						1044.5
U3 - Beta(1, 1)						1041.1
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	67.9	9.6	1	0.1	0	21.4
U2 - Beta(0.5, 0.5)	64.6	12.2	2.4	0.5	0.1	20.2
U2 - Beta(0.7, 0.3)	63.4	13.2	1.7	0.7	0.1	20.9
U2 - Beta(1, 1)	64.2	12.8	2.5	0.5	0.1	19.9
U3 - Beta(0.3, 0.7)	69.8	8.1	0.6	0.2	0	21.3
U3 - Beta(0.5, 0.5)	65.5	11.9	1.4	0.4	0.1	20.7
U3 - Beta(0.7, 0.3)	65.4	12.2	1.4	0.5	0.1	20.4
U3 - Beta(1, 1)	65.6	12	1.6	0.4	0.1	20.3
Mean administrations						
U2 - Beta(0.3, 0.7)	22.4	3.5	0.4	0.1	0	-
U2 - Beta(0.5, 0.5)	19	5.7	1.5	0.5	0.1	-
U2 - Beta(0.7, 0.3)	18.4	5.9	1.6	0.7	0.1	-
U2 - Beta(1, 1)	19	5.7	1.5	0.5	0.1	-
U3 - Beta(0.3, 0.7)	22.8	3.3	0.3	0.1	0	-
U3 - Beta(0.5, 0.5)	19.4	5.5	1.2	0.4	0.1	-
U3 - Beta(0.7, 0.3)	18.9	5.6	1.5	0.6	0.1	-
U3 - Beta(1, 1)	19.7	5.4	1.2	0.4	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	6.3	1.5	0.1	0	0	-
U2 - Beta(0.5, 0.5)	5	2.4	0.5	0.2	0	-
U2 - Beta(0.7, 0.3)	4.8	2.4	0.5	0.2	0	-
U2 - Beta(1, 1)	4.9	2.4	0.5	0.2	0	-
U3 - Beta(0.3, 0.7)	6.4	1.5	0.1	0	0	-
U3 - Beta(0.5, 0.5)	5.1	2.3	0.4	0.2	0	-
U3 - Beta(0.7, 0.3)	4.9	2.4	0.5	0.2	0	-
U3 - Beta(1, 1)	5.2	2.3	0.4	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1014.9
U2 - Beta(0.5, 0.5)						1033.8
U2 - Beta(0.7, 0.3)						1036.5
U2 - Beta(1, 1)						1036.4
U3 - Beta(0.3, 0.7)						1015
U3 - Beta(0.5, 0.5)						1032.7
U3 - Beta(0.7, 0.3)						1036.9
U3 - Beta(1, 1)						1036

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	21.6	69.2	4.7	0.3	0.1	4.1
U2 - Beta(0.5, 0.5)	15	57.6	20	3.9	0.5	3
U2 - Beta(0.7, 0.3)	15.3	49	23.2	8.5	1.2	2.8
U2 - Beta(1, 1)	14.6	57.9	20	3.8	0.5	3.2
U3 - Beta(0.3, 0.7)	25.1	67.5	2.9	0.3	0	4.2
U3 - Beta(0.5, 0.5)	16.4	61	15.5	3.3	0.5	3.3
U3 - Beta(0.7, 0.3)	15.7	52.8	21.3	6.2	0.8	3.2
U3 - Beta(1, 1)	15.9	61.6	15.4	3.6	0.4	3.1
Mean administrations						
U2 - Beta(0.3, 0.7)	12.8	15.4	1	0.1	0	-
U2 - Beta(0.5, 0.5)	8.9	14.7	4.7	1.1	0.1	-
U2 - Beta(0.7, 0.3)	8.6	11.1	5.9	3.2	0.8	-
U2 - Beta(1, 1)	8.7	14.7	4.8	1.2	0.1	-
U3 - Beta(0.3, 0.7)	13.5	15.2	0.7	0.1	0	-
U3 - Beta(0.5, 0.5)	9.2	15.3	4	0.9	0.1	-
U3 - Beta(0.7, 0.3)	8.8	12.4	5.6	2.4	0.3	-
U3 - Beta(1, 1)	9.2	15.4	4	0.9	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.7	5.4	0.3	0	0	-
U2 - Beta(0.5, 0.5)	1.9	5.3	1.3	0.4	0	-
U2 - Beta(0.7, 0.3)	1.8	4.2	1.6	0.9	0.3	-
U2 - Beta(1, 1)	1.8	5.3	1.4	0.4	0	-
U3 - Beta(0.3, 0.7)	3.9	5.3	0.2	0	0	-
U3 - Beta(0.5, 0.5)	2	5.5	1.1	0.3	0	-
U3 - Beta(0.7, 0.3)	1.9	4.6	1.6	0.8	0.1	-
U3 - Beta(1, 1)	2	5.5	1.1	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1102.5
U2 - Beta(0.5, 0.5)						1137.9
U2 - Beta(0.7, 0.3)						1144.7
U2 - Beta(1, 1)						1137.5
U3 - Beta(0.3, 0.7)						1099.7
U3 - Beta(0.5, 0.5)						1132.6
U3 - Beta(0.7, 0.3)						1139.8
U3 - Beta(1, 1)						1133.6
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	27.8	4.6	1.2	0.1	0	66.3
U2 - Beta(0.5, 0.5)	29.9	5	0.6	0.1	0	64.4
U2 - Beta(0.7, 0.3)	30.9	4.9	0.8	0.1	0	63.3
U2 - Beta(1, 1)	30.1	5.1	0.5	0.1	0	64.2
U3 - Beta(0.3, 0.7)	27.7	4.7	1	0	0	66.6
U3 - Beta(0.5, 0.5)	30.5	5.2	0.2	0.1	0	64
U3 - Beta(0.7, 0.3)	29.4	5.3	0.4	0.1	0	64.8
U3 - Beta(1, 1)	30.2	5	0.3	0.1	0	64.4
Mean administrations						
U2 - Beta(0.3, 0.7)	14.7	4.4	0.9	0.2	0	-
U2 - Beta(0.5, 0.5)	14.1	4.9	1.2	0.4	0.1	-
U2 - Beta(0.7, 0.3)	14.1	4.9	1.2	0.4	0.1	-
U2 - Beta(1, 1)	14	4.9	1.2	0.4	0.1	-
U3 - Beta(0.3, 0.7)	14.8	4.3	0.8	0.2	0	-
U3 - Beta(0.5, 0.5)	14.3	4.8	1.1	0.4	0.1	-
U3 - Beta(0.7, 0.3)	14.2	4.8	1.1	0.4	0.1	-
U3 - Beta(1, 1)	14.3	4.8	1.1	0.4	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.2	2.2	0.3	0.1	0	-
U2 - Beta(0.5, 0.5)	3.9	2.4	0.4	0.2	0	-
U2 - Beta(0.7, 0.3)	4	2.4	0.4	0.2	0	-
U2 - Beta(1, 1)	3.9	2.5	0.4	0.2	0	-
U3 - Beta(0.3, 0.7)	4.3	2.2	0.3	0.1	0	-
U3 - Beta(0.5, 0.5)	4	2.4	0.4	0.2	0	-
U3 - Beta(0.7, 0.3)	3.9	2.4	0.4	0.2	0	-
U3 - Beta(1, 1)	4	2.4	0.4	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						730.2
U2 - Beta(0.5, 0.5)						745.9
U2 - Beta(0.7, 0.3)						750.7
U2 - Beta(1, 1)						746.2
U3 - Beta(0.3, 0.7)						725.4
U3 - Beta(0.5, 0.5)						747.1
U3 - Beta(0.7, 0.3)						745.5
U3 - Beta(1, 1)						747.8

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.1	14.5	27.5	18.9	8.4	20.6
U2 - Beta(0.5, 0.5)	14.7	13.6	22.1	15.2	15.4	19
U2 - Beta(0.7, 0.3)	15.3	13.7	21.8	14.2	15.7	19.3
U2 - Beta(1, 1)	15.6	12.7	22.8	14.6	15.5	18.8
U3 - Beta(0.3, 0.7)	13.8	16.2	25.2	17.4	5.9	21.5
U3 - Beta(0.5, 0.5)	18.4	16.3	20.2	14.1	12.1	18.9
U3 - Beta(0.7, 0.3)	19.3	16.4	19	13.8	12.3	19.2
U3 - Beta(1, 1)	21.5	16.1	18.2	12.6	12.3	19.3
Mean administrations						
U2 - Beta(0.3, 0.7)	7.8	6.7	7.1	4.4	1.3	-
U2 - Beta(0.5, 0.5)	7.4	6.2	6	4.9	3.1	-
U2 - Beta(0.7, 0.3)	7.5	6.2	5.7	4.8	3.4	-
U2 - Beta(1, 1)	7.3	6.4	5.9	4.9	3.1	-
U3 - Beta(0.3, 0.7)	8.2	6.8	7.1	4.2	1	-
U3 - Beta(0.5, 0.5)	7.6	6.3	5.9	4.9	2.9	-
U3 - Beta(0.7, 0.3)	7.7	6.5	5.7	4.7	3	-
U3 - Beta(1, 1)	7.8	6.6	5.6	4.7	2.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.9	3.6	2.6	1.8	0.5	-
U2 - Beta(0.5, 0.5)	1.7	3.6	2	1.9	1	-
U2 - Beta(0.7, 0.3)	1.7	3.6	1.9	1.9	1.1	-
U2 - Beta(1, 1)	1.6	3.6	2	2	1	-
U3 - Beta(0.3, 0.7)	2.2	3.6	2.6	1.7	0.3	-
U3 - Beta(0.5, 0.5)	1.8	3.6	2	1.9	0.9	-
U3 - Beta(0.7, 0.3)	1.8	3.7	1.9	1.9	1	-
U3 - Beta(1, 1)	1.9	3.7	1.9	1.9	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						929.8
U2 - Beta(0.5, 0.5)						954.9
U2 - Beta(0.7, 0.3)						956.5
U2 - Beta(1, 1)						955.3
U3 - Beta(0.3, 0.7)						925.8
U3 - Beta(0.5, 0.5)						952.9
U3 - Beta(0.7, 0.3)						953.5
U3 - Beta(1, 1)						950.4
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.1	33.9	6.2	0.5	0	50.3
U2 - Beta(0.5, 0.5)	10.1	28.6	11.6	2.5	0.1	47.1
U2 - Beta(0.7, 0.3)	11.3	30.1	10.8	2.4	0.3	45.1
U2 - Beta(1, 1)	10.1	28.9	11.7	2.3	0.1	46.9
U3 - Beta(0.3, 0.7)	9.1	35.2	5.1	0.5	0	50.1
U3 - Beta(0.5, 0.5)	10.3	29.8	10.6	2.2	0.2	46.9
U3 - Beta(0.7, 0.3)	10.9	30.1	10.2	2.1	0.4	46.3
U3 - Beta(1, 1)	10.6	31.5	9.3	1.8	0.1	46.7
Mean administrations						
U2 - Beta(0.3, 0.7)	10.1	11.1	1.7	0.2	0	-
U2 - Beta(0.5, 0.5)	9.6	9.2	3.4	1.1	0.2	-
U2 - Beta(0.7, 0.3)	9.5	8.4	3.7	1.7	0.4	-
U2 - Beta(1, 1)	9.5	9.3	3.5	1.1	0.2	-
U3 - Beta(0.3, 0.7)	10.2	11.2	1.5	0.2	0	-
U3 - Beta(0.5, 0.5)	9.6	9.4	3.3	1	0.1	-
U3 - Beta(0.7, 0.3)	9.5	8.8	3.5	1.4	0.3	-
U3 - Beta(1, 1)	9.7	9.6	3.2	0.9	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	4.7	0.5	0.1	0	-
U2 - Beta(0.5, 0.5)	2.2	4.1	1	0.4	0.1	-
U2 - Beta(0.7, 0.3)	2.2	3.8	1	0.6	0.1	-
U2 - Beta(1, 1)	2.2	4.1	1	0.4	0.1	-
U3 - Beta(0.3, 0.7)	2.5	4.7	0.5	0.1	0	-
U3 - Beta(0.5, 0.5)	2.2	4.1	1	0.4	0.1	-
U3 - Beta(0.7, 0.3)	2.2	4	1	0.5	0.1	-
U3 - Beta(1, 1)	2.3	4.2	0.9	0.4	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						841.1
U2 - Beta(0.5, 0.5)						865.5
U2 - Beta(0.7, 0.3)						877.5
U2 - Beta(1, 1)						866.4
U3 - Beta(0.3, 0.7)						838.3
U3 - Beta(0.5, 0.5)						864.2
U3 - Beta(0.7, 0.3)						870.7
U3 - Beta(1, 1)						864.7

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	63.6	18.6	2.3	0	0	15.5
U2 - Beta(0.5, 0.5)	65.2	18	1.3	0	0	15.5
U2 - Beta(0.7, 0.3)	64.9	18.6	1.1	0	0	15.4
U2 - Beta(1, 1)	65.4	17.8	1.4	0	0	15.4
U3 - Beta(0.3, 0.7)	66	17	1.6	0	0	15.4
U3 - Beta(0.5, 0.5)	68	15.7	1.2	0	0	15.1
U3 - Beta(0.7, 0.3)	65.1	18.6	1.4	0.1	0	14.8
U3 - Beta(1, 1)	66.2	17.4	1	0.1	0	15.3
Mean administrations						
U2 - Beta(0.3, 0.7)	19.3	6.7	1.6	0.2	0	-
U2 - Beta(0.5, 0.5)	17.7	7.2	2.2	0.8	0.1	-
U2 - Beta(0.7, 0.3)	17.6	7.3	2.2	0.8	0.1	-
U2 - Beta(1, 1)	17.5	7.4	2.2	0.7	0	-
U3 - Beta(0.3, 0.7)	20	6.5	1.1	0.2	0	-
U3 - Beta(0.5, 0.5)	18.1	7	2	0.7	0	-
U3 - Beta(0.7, 0.3)	17.8	7.2	2.1	0.7	0	-
U3 - Beta(1, 1)	18	7.1	2	0.7	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.9	3.1	0.6	0.1	0	-
U2 - Beta(0.5, 0.5)	5.2	3.4	0.7	0.3	0	-
U2 - Beta(0.7, 0.3)	5.2	3.4	0.7	0.3	0	-
U2 - Beta(1, 1)	5.1	3.5	0.7	0.3	0	-
U3 - Beta(0.3, 0.7)	6.2	3	0.4	0.1	0	-
U3 - Beta(0.5, 0.5)	5.4	3.3	0.7	0.3	0	-
U3 - Beta(0.7, 0.3)	5.2	3.4	0.6	0.3	0	-
U3 - Beta(1, 1)	5.3	3.4	0.6	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						994.6
U2 - Beta(0.5, 0.5)						1006.7
U2 - Beta(0.7, 0.3)						1009.2
U2 - Beta(1, 1)						1006
U3 - Beta(0.3, 0.7)						993.5
U3 - Beta(0.5, 0.5)						1004.3
U3 - Beta(0.7, 0.3)						1007.5
U3 - Beta(1, 1)						1002.4

Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.4	14.5	14.5	55.1	4.7	2.8
U2 - Beta(0.5, 0.5)	10.8	15.3	13.3	45.8	12.4	2.4
U2 - Beta(0.7, 0.3)	14	15.5	12.3	43.3	12.5	2.4
U2 - Beta(1, 1)	14.1	14.2	11.6	44.9	12.7	2.5
U3 - Beta(0.3, 0.7)	12.7	16.8	11.9	52.5	3.2	2.9
U3 - Beta(0.5, 0.5)	16.6	15.7	10.2	44.3	10.9	2.3
U3 - Beta(0.7, 0.3)	18.1	17.7	10.3	39.3	12.2	2.4
U3 - Beta(1, 1)	17.9	16	10.8	41.8	10.9	2.6
Mean administrations						
U2 - Beta(0.3, 0.7)	4.9	6.5	7	10.1	1.2	-
U2 - Beta(0.5, 0.5)	4.7	5.9	6.2	9.4	3.6	-
U2 - Beta(0.7, 0.3)	4.8	5.9	6.1	8.5	4.5	-
U2 - Beta(1, 1)	4.8	5.9	6.1	9.3	3.6	-
U3 - Beta(0.3, 0.7)	5.3	7	6.8	9.6	1	-
U3 - Beta(0.5, 0.5)	5	6	6.1	9.2	3.4	-
U3 - Beta(0.7, 0.3)	5.3	6.2	5.8	8.5	4	-
U3 - Beta(1, 1)	5.4	6.1	5.9	9	3.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	3.7	2.5	3.6	0.4	-
U2 - Beta(0.5, 0.5)	0.7	3.6	1.9	3.3	1	-
U2 - Beta(0.7, 0.3)	0.8	3.6	1.8	3.1	1.1	-
U2 - Beta(1, 1)	0.8	3.6	1.8	3.3	1	-
U3 - Beta(0.3, 0.7)	1.1	3.8	2.4	3.5	0.3	-
U3 - Beta(0.5, 0.5)	0.9	3.7	1.8	3.3	0.9	-
U3 - Beta(0.7, 0.3)	1	3.8	1.7	3	1.1	-
U3 - Beta(1, 1)	1.1	3.7	1.7	3.2	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1032.1
U2 - Beta(0.5, 0.5)						1060.5
U2 - Beta(0.7, 0.3)						1065.2
U2 - Beta(1, 1)						1058.8
U3 - Beta(0.3, 0.7)						1024.1
U3 - Beta(0.5, 0.5)						1055.6
U3 - Beta(0.7, 0.3)						1057.8
U3 - Beta(1, 1)						1054.8

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.3	28.8	10.8	1.8	0.1	48.2
U2 - Beta(0.5, 0.5)	12.4	29.4	10	3.3	0.5	44.4
U2 - Beta(0.7, 0.3)	12.5	29.2	10.7	3.4	0.7	43.5
U2 - Beta(1, 1)	12.5	30.1	9.4	3.2	0.4	44.4
U3 - Beta(0.3, 0.7)	10.3	30.3	9	1.5	0.1	48.8
U3 - Beta(0.5, 0.5)	13.1	30.2	8.9	2.5	0.4	44.9
U3 - Beta(0.7, 0.3)	13.4	31	8.6	2.4	0.6	44
U3 - Beta(1, 1)	13.8	29.6	8.3	3.1	0.4	44.8
Mean administrations						
U2 - Beta(0.3, 0.7)	10	10.2	2.8	0.5	0	-
U2 - Beta(0.5, 0.5)	9.7	8.5	3.7	1.7	0.3	-
U2 - Beta(0.7, 0.3)	9.5	8.3	3.8	2	0.5	-
U2 - Beta(1, 1)	9.5	8.6	3.8	1.7	0.3	-
U3 - Beta(0.3, 0.7)	10.1	10.4	2.5	0.4	0	-
U3 - Beta(0.5, 0.5)	9.8	8.8	3.6	1.5	0.3	-
U3 - Beta(0.7, 0.3)	9.7	8.5	3.6	1.8	0.4	-
U3 - Beta(1, 1)	9.8	8.9	3.5	1.6	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	4.6	1	0.2	0	-
U2 - Beta(0.5, 0.5)	2.3	4	1.1	0.7	0.1	-
U2 - Beta(0.7, 0.3)	2.3	4	1.1	0.7	0.2	-
U2 - Beta(1, 1)	2.3	4.1	1.2	0.7	0.1	-
U3 - Beta(0.3, 0.7)	2.6	4.7	0.9	0.2	0	-
U3 - Beta(0.5, 0.5)	2.4	4.2	1.1	0.6	0.1	-
U3 - Beta(0.7, 0.3)	2.4	4.1	1.1	0.7	0.1	-
U3 - Beta(1, 1)	2.4	4.2	1.1	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						833.9
U2 - Beta(0.5, 0.5)						857.5
U2 - Beta(0.7, 0.3)						865
U2 - Beta(1, 1)						857.6
U3 - Beta(0.3, 0.7)						832.7
U3 - Beta(0.5, 0.5)						855.9
U3 - Beta(0.7, 0.3)						861.5
U3 - Beta(1, 1)						855.8

Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.6	17.4	18.9	34.6	7.1	8.4
U2 - Beta(0.5, 0.5)	17.5	19.6	18	28	8.6	8.3
U2 - Beta(0.7, 0.3)	18.6	18.6	18.7	26.5	9.4	8.2
U2 - Beta(1, 1)	17.8	18.1	17.7	29.3	8.7	8.4
U3 - Beta(0.3, 0.7)	18.2	20.3	15.6	31.9	5.2	8.8
U3 - Beta(0.5, 0.5)	22.3	21.3	15	26.2	7	8.2
U3 - Beta(0.7, 0.3)	23.4	22.7	15.1	23.7	6.5	8.6
U3 - Beta(1, 1)	23.5	21.2	15.1	25.4	6.4	8.4
Mean administrations						
U2 - Beta(0.3, 0.7)	6.4	7.2	7.1	7	1.5	-
U2 - Beta(0.5, 0.5)	6.2	6.8	6.4	6.9	2.9	-
U2 - Beta(0.7, 0.3)	6.3	6.8	6.3	6.6	3.2	-
U2 - Beta(1, 1)	6.3	6.8	6.3	6.9	2.9	-
U3 - Beta(0.3, 0.7)	7	7.6	6.6	6.7	1.2	-
U3 - Beta(0.5, 0.5)	6.7	6.9	6.1	6.8	2.7	-
U3 - Beta(0.7, 0.3)	6.8	7.1	6	6.5	2.9	-
U3 - Beta(1, 1)	6.9	7.1	6	6.5	2.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.5	3.9	2.6	2.7	0.5	-
U2 - Beta(0.5, 0.5)	1.4	3.9	2.1	2.7	1	-
U2 - Beta(0.7, 0.3)	1.4	3.9	2	2.6	1	-
U2 - Beta(1, 1)	1.4	4	2	2.7	1	-
U3 - Beta(0.3, 0.7)	1.8	4.1	2.4	2.6	0.5	-
U3 - Beta(0.5, 0.5)	1.6	4	1.9	2.6	0.9	-
U3 - Beta(0.7, 0.3)	1.6	4	1.9	2.5	1	-
U3 - Beta(1, 1)	1.7	4.1	1.9	2.5	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						986.4
U2 - Beta(0.5, 0.5)						1000.2
U2 - Beta(0.7, 0.3)						1001.6
U2 - Beta(1, 1)						1000.7
U3 - Beta(0.3, 0.7)						981.6
U3 - Beta(0.5, 0.5)						998.5
U3 - Beta(0.7, 0.3)						997.8
U3 - Beta(1, 1)						995.9

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	37.5	32	12.5	3	0.1	14.9
U2 - Beta(0.5, 0.5)	41.9	29.3	10.9	3.5	0.2	14.2
U2 - Beta(0.7, 0.3)	41.5	29	11.5	3.9	0.3	13.8
U2 - Beta(1, 1)	41.1	29.3	11.7	4.4	0.3	13.2
U3 - Beta(0.3, 0.7)	44	31.4	8.2	1.7	0.1	14.6
U3 - Beta(0.5, 0.5)	45.5	30.5	6.7	2.9	0.3	14.1
U3 - Beta(0.7, 0.3)	44.1	31.5	7.6	2.8	0.3	13.7
U3 - Beta(1, 1)	42.4	31.2	8.9	3	0.2	14.3
Mean administrations						
U2 - Beta(0.3, 0.7)	10.6	9.1	5.5	2.6	0.6	-
U2 - Beta(0.5, 0.5)	10.6	8.7	5.4	3	0.8	-
U2 - Beta(0.7, 0.3)	10.7	8.6	5.3	3.1	0.9	-
U2 - Beta(1, 1)	10.5	8.8	5.3	3.1	0.8	-
U3 - Beta(0.3, 0.7)	11.9	9	4.7	2.3	0.5	-
U3 - Beta(0.5, 0.5)	11.1	8.6	5	2.9	0.8	-
U3 - Beta(0.7, 0.3)	11.2	8.6	4.9	3	0.8	-
U3 - Beta(1, 1)	11.3	8.6	4.9	2.9	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	4.4	2.1	1.1	0.3	-
U2 - Beta(0.5, 0.5)	3.1	4.5	1.8	1.3	0.3	-
U2 - Beta(0.7, 0.3)	3.1	4.4	1.8	1.4	0.3	-
U2 - Beta(1, 1)	3	4.5	1.8	1.3	0.3	-
U3 - Beta(0.3, 0.7)	3.8	4.4	1.8	0.9	0.2	-
U3 - Beta(0.5, 0.5)	3.3	4.4	1.7	1.2	0.3	-
U3 - Beta(0.7, 0.3)	3.4	4.4	1.7	1.3	0.3	-
U3 - Beta(1, 1)	3.4	4.4	1.6	1.3	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						953.4
U2 - Beta(0.5, 0.5)						961.2
U2 - Beta(0.7, 0.3)						960.7
U2 - Beta(1, 1)						962.8
U3 - Beta(0.3, 0.7)						948.6
U3 - Beta(0.5, 0.5)						959
U3 - Beta(0.7, 0.3)						959.2
U3 - Beta(1, 1)						957.1
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	21.2	25.4	20.3	20.8	12.3	0
U2 - Beta(0.5, 0.5)	36.1	29.3	14.1	13.2	7.3	0
U2 - Beta(0.7, 0.3)	36.4	32	14.5	11.4	5.7	0
U2 - Beta(1, 1)	37.9	30	15.4	10.6	6.1	0
U3 - Beta(0.3, 0.7)	28.3	29.7	20.6	15	6.4	0
U3 - Beta(0.5, 0.5)	40.5	32.2	12.7	10.2	4.4	0
U3 - Beta(0.7, 0.3)	38.6	33.4	14.8	9.4	3.8	0
U3 - Beta(1, 1)	44.2	31.2	12.9	8.4	3.3	0
Mean administrations						
U2 - Beta(0.3, 0.7)	9.2	8.2	5.7	4.7	2.2	-
U2 - Beta(0.5, 0.5)	4.3	5.3	6.1	8.3	6	-
U2 - Beta(0.7, 0.3)	4.2	5.1	6	8.4	6.2	-
U2 - Beta(1, 1)	4.4	5.5	6.2	8.1	5.8	-
U3 - Beta(0.3, 0.7)	9.8	8.9	5.5	4.1	1.6	-
U3 - Beta(0.5, 0.5)	4.8	5.8	6.1	8	5.3	-
U3 - Beta(0.7, 0.3)	4.9	5.7	6.3	8	5.2	-
U3 - Beta(1, 1)	4.9	6	6.3	7.8	5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	4	2.4	1.9	0.9	-
U2 - Beta(0.5, 0.5)	0.5	3	2.3	3.5	2.3	-
U2 - Beta(0.7, 0.3)	0.5	3	2.1	3.7	2.2	-
U2 - Beta(1, 1)	0.5	3.1	2.4	3.4	2.2	-
U3 - Beta(0.3, 0.7)	3	4.2	2.4	1.7	0.7	-
U3 - Beta(0.5, 0.5)	0.7	3.2	2.3	3.4	2	-
U3 - Beta(0.7, 0.3)	0.8	3.2	2.3	3.4	1.9	-
U3 - Beta(1, 1)	0.8	3.3	2.4	3.3	1.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						988.8
U2 - Beta(0.5, 0.5)						1007.9
U2 - Beta(0.7, 0.3)						1011.3
U2 - Beta(1, 1)						1007.4
U3 - Beta(0.3, 0.7)						986.4
U3 - Beta(0.5, 0.5)						1005.5
U3 - Beta(0.7, 0.3)						1007
U3 - Beta(1, 1)						1003.9

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	48.6	48.4	2.2	0.2	0	0.6
U2 - Beta(0.5, 0.5)	21.8	68.2	9.1	0.4	0	0.5
U2 - Beta(0.7, 0.3)	22.3	61.7	13.9	1.5	0	0.6
U2 - Beta(1, 1)	20.3	68.2	10.5	0.5	0	0.5
U3 - Beta(0.3, 0.7)	51.7	46.4	1.2	0	0	0.7
U3 - Beta(0.5, 0.5)	23.5	67.9	7.8	0.3	0	0.5
U3 - Beta(0.7, 0.3)	24.4	62.4	11.4	1	0	0.8
U3 - Beta(1, 1)	23.6	68.2	7.4	0.3	0	0.5
Mean administrations						
U2 - Beta(0.3, 0.7)	18.7	10.6	0.6	0	0	-
U2 - Beta(0.5, 0.5)	10.1	16.2	3.2	0.3	0	-
U2 - Beta(0.7, 0.3)	9.6	13.6	5.1	1.5	0.1	-
U2 - Beta(1, 1)	9.9	16.2	3.4	0.4	0	-
U3 - Beta(0.3, 0.7)	19.2	10.4	0.3	0	0	-
U3 - Beta(0.5, 0.5)	10.8	16.3	2.6	0.2	0	-
U3 - Beta(0.7, 0.3)	10	14.5	4.4	1	0.1	-
U3 - Beta(1, 1)	10.7	16.4	2.5	0.2	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.8	3.7	0.2	0	0	-
U2 - Beta(0.5, 0.5)	2.3	5.4	0.9	0.1	0	-
U2 - Beta(0.7, 0.3)	2.1	4.6	1.3	0.4	0	-
U2 - Beta(1, 1)	2.2	5.4	1	0.1	0	-
U3 - Beta(0.3, 0.7)	6	3.6	0.1	0	0	-
U3 - Beta(0.5, 0.5)	2.5	5.5	0.7	0.1	0	-
U3 - Beta(0.7, 0.3)	2.2	4.9	1.2	0.3	0	-
U3 - Beta(1, 1)	2.5	5.5	0.7	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1117.1
U2 - Beta(0.5, 0.5)						1170
U2 - Beta(0.7, 0.3)						1186.6
U2 - Beta(1, 1)						1171.3
U3 - Beta(0.3, 0.7)						1114.3
U3 - Beta(0.5, 0.5)						1164.7
U3 - Beta(0.7, 0.3)						1175.2
U3 - Beta(1, 1)						1165.4
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.5	8.7	12.6	30.6	44.5	0.1
U2 - Beta(0.5, 0.5)	8.7	12.1	10.6	19.3	49.2	0.1
U2 - Beta(0.7, 0.3)	10.1	13.6	10.8	18.1	47.3	0.1
U2 - Beta(1, 1)	12.2	10.7	10	17.6	49.4	0.1
U3 - Beta(0.3, 0.7)	9.1	10.8	15.8	29.1	35.1	0.1
U3 - Beta(0.5, 0.5)	15.4	12.6	11.3	15.9	44.7	0.1
U3 - Beta(0.7, 0.3)	17.6	15.3	12.5	13.8	40.7	0.1
U3 - Beta(1, 1)	19	15.6	10.2	12.3	42.8	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.8	5.2	6.5	8.4	6.1	-
U2 - Beta(0.5, 0.5)	3.4	4.1	5.3	7.4	9.7	-
U2 - Beta(0.7, 0.3)	3.5	4.1	5.3	7.2	9.9	-
U2 - Beta(1, 1)	3.5	4.2	5.4	7.3	9.7	-
U3 - Beta(0.3, 0.7)	4	5.5	6.9	8.3	5.2	-
U3 - Beta(0.5, 0.5)	3.5	4.2	5.5	7.6	9.2	-
U3 - Beta(0.7, 0.3)	3.7	4.4	5.3	7.3	9.2	-
U3 - Beta(1, 1)	3.7	4.5	5.4	7.4	8.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.5	3.1	2.7	3.3	2	-
U2 - Beta(0.5, 0.5)	0.2	2.8	1.8	3	2.9	-
U2 - Beta(0.7, 0.3)	0.2	2.9	1.8	3	2.8	-
U2 - Beta(1, 1)	0.2	2.9	1.8	3	2.9	-
U3 - Beta(0.3, 0.7)	0.5	3.3	2.8	3.3	1.7	-
U3 - Beta(0.5, 0.5)	0.3	2.9	1.9	3.1	2.7	-
U3 - Beta(0.7, 0.3)	0.3	3	1.8	3	2.7	-
U3 - Beta(1, 1)	0.4	3	1.9	3	2.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1017.8
U2 - Beta(0.5, 0.5)						1057.3
U2 - Beta(0.7, 0.3)						1060.4
U2 - Beta(1, 1)						1055.6
U3 - Beta(0.3, 0.7)						1010.3
U3 - Beta(0.5, 0.5)						1052.2
U3 - Beta(0.7, 0.3)						1053.4
U3 - Beta(1, 1)						1049.1

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.4	58.8	12.2	0.4	0	15.2
U2 - Beta(0.5, 0.5)	16.3	54.2	13.4	0.4	0	15.7
U2 - Beta(0.7, 0.3)	16.3	56.2	12	0.7	0	14.8
U2 - Beta(1, 1)	15.9	55.6	12.7	0.5	0	15.3
U3 - Beta(0.3, 0.7)	14.8	60.7	9.1	0.3	0	15.1
U3 - Beta(0.5, 0.5)	18.2	54.8	11.2	0.2	0	15.6
U3 - Beta(0.7, 0.3)	17.4	56	11.6	0.6	0	14.4
U3 - Beta(1, 1)	17.7	55.3	11.2	0.3	0	15.5
Mean administrations						
U2 - Beta(0.3, 0.7)	8.9	15.3	3.7	0.4	0	-
U2 - Beta(0.5, 0.5)	8.7	12.8	5.3	1.4	0.1	-
U2 - Beta(0.7, 0.3)	8.7	12.4	5.3	1.9	0.2	-
U2 - Beta(1, 1)	8.6	13	5.3	1.4	0.1	-
U3 - Beta(0.3, 0.7)	9.4	15.4	3.2	0.3	0	-
U3 - Beta(0.5, 0.5)	9	13.1	5	1.2	0.1	-
U3 - Beta(0.7, 0.3)	8.9	12.7	5	1.6	0.1	-
U3 - Beta(1, 1)	8.9	13.2	4.9	1.3	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.2	6.4	1.3	0.2	0	-
U2 - Beta(0.5, 0.5)	2	5.6	1.6	0.6	0	-
U2 - Beta(0.7, 0.3)	2	5.5	1.5	0.7	0.1	-
U2 - Beta(1, 1)	2	5.6	1.6	0.6	0	-
U3 - Beta(0.3, 0.7)	2.4	6.4	1.1	0.1	0	-
U3 - Beta(0.5, 0.5)	2.1	5.7	1.5	0.5	0	-
U3 - Beta(0.7, 0.3)	2.1	5.6	1.5	0.6	0.1	-
U3 - Beta(1, 1)	2.1	5.8	1.5	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1004
U2 - Beta(0.5, 0.5)						1017.5
U2 - Beta(0.7, 0.3)						1026.4
U2 - Beta(1, 1)						1019.4
U3 - Beta(0.3, 0.7)						996.8
U3 - Beta(0.5, 0.5)						1013.1
U3 - Beta(0.7, 0.3)						1021.1
U3 - Beta(1, 1)						1014.3

Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6	53.2	30.7	6.6	0.6	2.9
U2 - Beta(0.5, 0.5)	8.7	52.4	28.8	7	0.7	2.4
U2 - Beta(0.7, 0.3)	10.7	50	28	8.2	0.6	2.5
U2 - Beta(1, 1)	9.9	52.4	26.4	8.3	0.6	2.4
U3 - Beta(0.3, 0.7)	9	57	25.6	5.4	0.2	2.8
U3 - Beta(0.5, 0.5)	12.5	55.9	22.3	6.4	0.5	2.4
U3 - Beta(0.7, 0.3)	11.8	55	23	7.3	0.4	2.5
U3 - Beta(1, 1)	12.9	56.3	20	7.6	0.5	2.7
Mean administrations						
U2 - Beta(0.3, 0.7)	5.2	13.9	7.8	2.4	0.3	-
U2 - Beta(0.5, 0.5)	5.1	11.1	8.2	4.2	1	-
U2 - Beta(0.7, 0.3)	5.2	10.7	8.1	4.4	1.2	-
U2 - Beta(1, 1)	5.2	11.1	8.2	4.1	1	-
U3 - Beta(0.3, 0.7)	5.5	14.9	7.1	1.9	0.2	-
U3 - Beta(0.5, 0.5)	5.4	11.7	7.8	3.9	0.9	-
U3 - Beta(0.7, 0.3)	5.5	11.3	7.6	4.2	1	-
U3 - Beta(1, 1)	5.6	11.9	7.5	3.8	0.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	6.2	2.8	1	0.1	-
U2 - Beta(0.5, 0.5)	0.8	5.2	2.6	1.6	0.4	-
U2 - Beta(0.7, 0.3)	0.9	5.2	2.5	1.7	0.4	-
U2 - Beta(1, 1)	0.9	5.2	2.6	1.6	0.4	-
U3 - Beta(0.3, 0.7)	1.1	6.6	2.5	0.8	0.1	-
U3 - Beta(0.5, 0.5)	0.9	5.5	2.5	1.5	0.3	-
U3 - Beta(0.7, 0.3)	1	5.3	2.4	1.5	0.4	-
U3 - Beta(1, 1)	1	5.6	2.4	1.5	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1025.7
U2 - Beta(0.5, 0.5)						1045
U2 - Beta(0.7, 0.3)						1046.7
U2 - Beta(1, 1)						1044
U3 - Beta(0.3, 0.7)						1021.1
U3 - Beta(0.5, 0.5)						1041
U3 - Beta(0.7, 0.3)						1044
U3 - Beta(1, 1)						1038.9

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	48.8	0.1	0	0	0	51.1
U2 - Beta(0.5, 0.5)	45.4	0.2	0	0	0	54.4
U2 - Beta(0.7, 0.3)	42.1	0.7	0	0	0	57.2
U2 - Beta(1, 1)	45.2	0.1	0	0	0	54.7
U3 - Beta(0.3, 0.7)	48.7	0	0	0	0	51.3
U3 - Beta(0.5, 0.5)	45.3	0.1	0	0	0	54.6
U3 - Beta(0.7, 0.3)	44.3	0.4	0	0	0	55.3
U3 - Beta(1, 1)	45.4	0.1	0	0	0	54.5
Mean administrations						
U2 - Beta(0.3, 0.7)	21.3	0.4	0	0	0	-
U2 - Beta(0.5, 0.5)	19.2	1.9	0	0	0	-
U2 - Beta(0.7, 0.3)	17.6	3.1	0.1	0	0	-
U2 - Beta(1, 1)	19.1	1.9	0	0	0	-
U3 - Beta(0.3, 0.7)	21.3	0.4	0	0	0	-
U3 - Beta(0.5, 0.5)	19.2	1.9	0	0	0	-
U3 - Beta(0.7, 0.3)	18.5	2.5	0.1	0	0	-
U3 - Beta(1, 1)	19.3	1.8	0	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.5	0.2	0	0	0	-
U2 - Beta(0.5, 0.5)	3.7	0.8	0	0	0	-
U2 - Beta(0.7, 0.3)	3.3	1.1	0	0	0	-
U2 - Beta(1, 1)	3.7	0.8	0	0	0	-
U3 - Beta(0.3, 0.7)	4.5	0.2	0	0	0	-
U3 - Beta(0.5, 0.5)	3.7	0.8	0	0	0	-
U3 - Beta(0.7, 0.3)	3.5	1	0	0	0	-
U3 - Beta(1, 1)	3.7	0.8	0	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						931.1
U2 - Beta(0.5, 0.5)						916.4
U2 - Beta(0.7, 0.3)						904.3
U2 - Beta(1, 1)						910.8
U3 - Beta(0.3, 0.7)						932
U3 - Beta(0.5, 0.5)						915.1
U3 - Beta(0.7, 0.3)						913.9
U3 - Beta(1, 1)						915.1
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	17.1	62.6	8.4	0.5	0	11.4
U2 - Beta(0.5, 0.5)	18.3	61.1	7	0.9	0	12.7
U2 - Beta(0.7, 0.3)	19.1	58.5	8.5	1.1	0	12.8
U2 - Beta(1, 1)	18.6	60	7.9	1.1	0	12.4
U3 - Beta(0.3, 0.7)	19	63.3	5.9	0.4	0	11.4
U3 - Beta(0.5, 0.5)	19.4	60.5	6.5	1.1	0	12.5
U3 - Beta(0.7, 0.3)	19.8	58.3	7.3	1.4	0	13.2
U3 - Beta(1, 1)	20.1	59	7.2	1	0	12.7
Mean administrations						
U2 - Beta(0.3, 0.7)	8.7	15.8	3.8	0.6	0	-
U2 - Beta(0.5, 0.5)	8.9	13.7	4.8	1.4	0.1	-
U2 - Beta(0.7, 0.3)	8.9	13.2	4.9	1.7	0.1	-
U2 - Beta(1, 1)	8.8	13.6	4.9	1.4	0	-
U3 - Beta(0.3, 0.7)	9.3	15.9	3.2	0.5	0	-
U3 - Beta(0.5, 0.5)	9.2	13.6	4.7	1.4	0	-
U3 - Beta(0.7, 0.3)	9.2	13.3	4.8	1.6	0.1	-
U3 - Beta(1, 1)	9.3	13.6	4.6	1.4	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.2	6.6	1.4	0.2	0	-
U2 - Beta(0.5, 0.5)	2.1	6.1	1.4	0.6	0	-
U2 - Beta(0.7, 0.3)	2.2	6	1.4	0.7	0	-
U2 - Beta(1, 1)	2.1	6.1	1.5	0.6	0	-
U3 - Beta(0.3, 0.7)	2.5	6.7	1.2	0.2	0	-
U3 - Beta(0.5, 0.5)	2.3	6.1	1.4	0.5	0	-
U3 - Beta(0.7, 0.3)	2.3	6	1.4	0.6	0	-
U3 - Beta(1, 1)	2.3	6.1	1.4	0.5	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1018.1
U2 - Beta(0.5, 0.5)						1023.5
U2 - Beta(0.7, 0.3)						1028.2
U2 - Beta(1, 1)						1025.1
U3 - Beta(0.3, 0.7)						1009.5
U3 - Beta(0.5, 0.5)						1022.4
U3 - Beta(0.7, 0.3)						1024.8
U3 - Beta(1, 1)						1021.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.7	0.1	0	0	0	93.2
U2 - Beta(0.5, 0.5)	6.7	0.4	0	0	0	92.9
U2 - Beta(0.7, 0.3)	6.5	0.5	0	0	0	93
U2 - Beta(1, 1)	6.4	0.6	0	0	0	93
U3 - Beta(0.3, 0.7)	6.4	0.1	0	0	0	93.5
U3 - Beta(0.5, 0.5)	6.5	0.4	0	0	0	93.1
U3 - Beta(0.7, 0.3)	6.5	0.3	0	0	0	93.2
U3 - Beta(1, 1)	6.4	0.3	0	0	0	93.3
Mean administrations						
U2 - Beta(0.3, 0.7)	11.1	3.3	0.3	0	0	-
U2 - Beta(0.5, 0.5)	10.9	3.5	0.3	0	0	-
U2 - Beta(0.7, 0.3)	10.9	3.5	0.3	0	0	-
U2 - Beta(1, 1)	10.9	3.5	0.3	0	0	-
U3 - Beta(0.3, 0.7)	11.2	3.3	0.3	0	0	-
U3 - Beta(0.5, 0.5)	10.9	3.5	0.3	0	0	-
U3 - Beta(0.7, 0.3)	10.9	3.5	0.3	0	0	-
U3 - Beta(1, 1)	10.9	3.5	0.3	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	1.9	0.1	0	0	-
U2 - Beta(0.5, 0.5)	2.9	2	0.1	0	0	-
U2 - Beta(0.7, 0.3)	2.8	2	0.1	0	0	-
U2 - Beta(1, 1)	2.9	2	0.1	0	0	-
U3 - Beta(0.3, 0.7)	3	1.9	0.1	0	0	-
U3 - Beta(0.5, 0.5)	2.9	2	0.1	0	0	-
U3 - Beta(0.7, 0.3)	2.9	2	0.1	0	0	-
U3 - Beta(1, 1)	2.8	2	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						532.1
U2 - Beta(0.5, 0.5)						529.4
U2 - Beta(0.7, 0.3)						528
U2 - Beta(1, 1)						529.6
U3 - Beta(0.3, 0.7)						531.2
U3 - Beta(0.5, 0.5)						528.8
U3 - Beta(0.7, 0.3)						529.1
U3 - Beta(1, 1)						526.4
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	0.3	0.6	12.1	74.7	12.3	0
U2 - Beta(0.5, 0.5)	0.9	0.8	2.1	60.8	35.4	0
U2 - Beta(0.7, 0.3)	2	2.5	4.7	51.1	39.7	0
U2 - Beta(1, 1)	1.1	0.7	2.2	60.5	35.5	0
U3 - Beta(0.3, 0.7)	0.3	1.7	14.3	75.1	8.6	0
U3 - Beta(0.5, 0.5)	1.3	1.5	2.5	63.6	31.1	0
U3 - Beta(0.7, 0.3)	2.6	3.4	4	52.8	37.2	0
U3 - Beta(1, 1)	1.9	1.9	2.7	64.4	29.1	0
Mean administrations						
U2 - Beta(0.3, 0.7)	3.6	4.3	7.4	13.1	1.6	-
U2 - Beta(0.5, 0.5)	3.1	3.2	3.8	13.5	6.5	-
U2 - Beta(0.7, 0.3)	3	3.1	3.7	11.2	8.9	-
U2 - Beta(1, 1)	3.1	3.2	3.8	13.5	6.4	-
U3 - Beta(0.3, 0.7)	3.6	4.4	7.7	13.1	1.2	-
U3 - Beta(0.5, 0.5)	3.1	3.2	3.9	14	5.9	-
U3 - Beta(0.7, 0.3)	3.1	3.2	3.9	11.7	8.1	-
U3 - Beta(1, 1)	3.1	3.2	3.9	14	5.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.3	2.7	3	4.5	0.5	-
U2 - Beta(0.5, 0.5)	0	2.5	1.2	4.7	2	-
U2 - Beta(0.7, 0.3)	0	2.4	1.1	4	2.5	-
U2 - Beta(1, 1)	0	2.4	1.2	4.6	2	-
U3 - Beta(0.3, 0.7)	0.4	2.8	3.1	4.6	0.4	-
U3 - Beta(0.5, 0.5)	0	2.5	1.2	4.8	1.8	-
U3 - Beta(0.7, 0.3)	0	2.5	1.2	4.1	2.4	-
U3 - Beta(1, 1)	0	2.5	1.2	4.8	1.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1040.7
U2 - Beta(0.5, 0.5)						1088.2
U2 - Beta(0.7, 0.3)						1096
U2 - Beta(1, 1)						1088.9
U3 - Beta(0.3, 0.7)						1038.1
U3 - Beta(0.5, 0.5)						1087
U3 - Beta(0.7, 0.3)						1090.2
U3 - Beta(1, 1)						1087.5

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	27.7	43.9	25.8	1.5	0	1.1
U2 - Beta(0.5, 0.5)	27.6	38	31.6	1.8	0	1
U2 - Beta(0.7, 0.3)	30.1	38.4	27.3	2.9	0	1.3
U2 - Beta(1, 1)	29	36.4	31.6	2.1	0	0.9
U3 - Beta(0.3, 0.7)	34.7	46.7	16.5	1	0	1.1
U3 - Beta(0.5, 0.5)	33.3	37.7	26.4	1.4	0	1.2
U3 - Beta(0.7, 0.3)	30.8	40.4	26.1	1.9	0	0.8
U3 - Beta(1, 1)	32.6	38.2	26.4	2	0	0.8
Mean administrations						
U2 - Beta(0.3, 0.7)	11.7	12	5.7	0.5	0	-
U2 - Beta(0.5, 0.5)	9.2	10.6	8.1	1.8	0.1	-
U2 - Beta(0.7, 0.3)	9.6	10.3	7.3	2.5	0.2	-
U2 - Beta(1, 1)	9.2	10.6	8.2	1.7	0.1	-
U3 - Beta(0.3, 0.7)	13.1	12.3	4.1	0.3	0	-
U3 - Beta(0.5, 0.5)	9.8	10.8	7.7	1.5	0.1	-
U3 - Beta(0.7, 0.3)	9.7	10.6	7.2	2.2	0.1	-
U3 - Beta(1, 1)	10	10.9	7.4	1.5	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.5	5	1.9	0.2	0	-
U2 - Beta(0.5, 0.5)	2.4	4.5	2.4	0.6	0	-
U2 - Beta(0.7, 0.3)	2.5	4.5	1.9	0.8	0.1	-
U2 - Beta(1, 1)	2.4	4.5	2.4	0.6	0	-
U3 - Beta(0.3, 0.7)	4.1	5.2	1.4	0.1	0	-
U3 - Beta(0.5, 0.5)	2.6	4.6	2.2	0.6	0	-
U3 - Beta(0.7, 0.3)	2.6	4.6	2	0.8	0	-
U3 - Beta(1, 1)	2.7	4.7	2.1	0.6	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1063.2
U2 - Beta(0.5, 0.5)						1099.3
U2 - Beta(0.7, 0.3)						1104.6
U2 - Beta(1, 1)						1100.6
U3 - Beta(0.3, 0.7)						1051.2
U3 - Beta(0.5, 0.5)						1094.7
U3 - Beta(0.7, 0.3)						1101.2
U3 - Beta(1, 1)						1091.8

Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	20.3	34.9	30.1	1.1	0	13.6
U2 - Beta(0.5, 0.5)	23.2	34.8	25.4	3.2	0	13.4
U2 - Beta(0.7, 0.3)	22.4	37.3	23.2	3.5	0	13.6
U2 - Beta(1, 1)	22.4	34.4	26.9	3	0	13.3
U3 - Beta(0.3, 0.7)	23.5	37.8	23.6	1.2	0	13.9
U3 - Beta(0.5, 0.5)	25.1	36.9	22.6	2.5	0	12.9
U3 - Beta(0.7, 0.3)	25.4	35.9	21.9	2.6	0	14.2
U3 - Beta(1, 1)	25.5	36	22.1	2.8	0	13.6
Mean administrations						
U2 - Beta(0.3, 0.7)	9.8	10.9	6.8	0.8	0	-
U2 - Beta(0.5, 0.5)	9.5	9.7	7	2.1	0.1	-
U2 - Beta(0.7, 0.3)	9.7	9.8	6.3	2.4	0.2	-
U2 - Beta(1, 1)	9.2	9.7	7.1	2.1	0.1	-
U3 - Beta(0.3, 0.7)	10.8	11.1	5.7	0.5	0	-
U3 - Beta(0.5, 0.5)	9.8	9.8	6.7	1.9	0.1	-
U3 - Beta(0.7, 0.3)	9.8	9.6	6.4	2.3	0.1	-
U3 - Beta(1, 1)	9.7	9.9	6.7	1.9	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.6	4.9	2.3	0.3	0	-
U2 - Beta(0.5, 0.5)	2.4	4.6	2.1	0.8	0	-
U2 - Beta(0.7, 0.3)	2.5	4.6	1.8	0.9	0.1	-
U2 - Beta(1, 1)	2.3	4.6	2.1	0.8	0	-
U3 - Beta(0.3, 0.7)	3.1	5	2	0.2	0	-
U3 - Beta(0.5, 0.5)	2.6	4.6	2	0.8	0	-
U3 - Beta(0.7, 0.3)	2.5	4.6	1.8	0.9	0	-
U3 - Beta(1, 1)	2.5	4.7	2	0.8	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						989.8
U2 - Beta(0.5, 0.5)						1012.6
U2 - Beta(0.7, 0.3)						1016.5
U2 - Beta(1, 1)						1012.2
U3 - Beta(0.3, 0.7)						981.7
U3 - Beta(0.5, 0.5)						1009.7
U3 - Beta(0.7, 0.3)						1011.8
U3 - Beta(1, 1)						1005.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.1	32.8	47.8	2.8	0.1	5.4
U2 - Beta(0.5, 0.5)	14.5	29.3	43	7	0.1	6.1
U2 - Beta(0.7, 0.3)	15.5	31	41.6	6.3	0	5.6
U2 - Beta(1, 1)	14.6	29.8	43.9	6.3	0.1	5.3
U3 - Beta(0.3, 0.7)	17	36.1	39	2.4	0.1	5.4
U3 - Beta(0.5, 0.5)	17.8	31.2	40.4	4.9	0.1	5.6
U3 - Beta(0.7, 0.3)	18.1	30.6	39	6.7	0.1	5.5
U3 - Beta(1, 1)	18.7	30.7	39.3	5.5	0.1	5.7
Mean administrations						
U2 - Beta(0.3, 0.7)	7.5	11.2	9.6	1	0	-
U2 - Beta(0.5, 0.5)	7	9.2	9.8	2.9	0.4	-
U2 - Beta(0.7, 0.3)	7.3	9	8.8	3.6	0.7	-
U2 - Beta(1, 1)	7.1	9.2	9.8	2.9	0.4	-
U3 - Beta(0.3, 0.7)	8.6	11.6	8.4	0.7	0	-
U3 - Beta(0.5, 0.5)	7.3	9.4	9.6	2.7	0.3	-
U3 - Beta(0.7, 0.3)	7.5	9.2	8.8	3.4	0.4	-
U3 - Beta(1, 1)	7.5	9.5	9.5	2.6	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.9	5	3.2	0.4	0	-
U2 - Beta(0.5, 0.5)	1.5	4.3	2.9	1.1	0.1	-
U2 - Beta(0.7, 0.3)	1.7	4.3	2.5	1.2	0.2	-
U2 - Beta(1, 1)	1.5	4.3	2.9	1.1	0.1	-
U3 - Beta(0.3, 0.7)	2.4	5.2	2.8	0.3	0	-
U3 - Beta(0.5, 0.5)	1.7	4.4	2.8	1	0.1	-
U3 - Beta(0.7, 0.3)	1.8	4.3	2.5	1.1	0.1	-
U3 - Beta(1, 1)	1.8	4.4	2.8	1	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1036.2
U2 - Beta(0.5, 0.5)						1062.5
U2 - Beta(0.7, 0.3)						1067.3
U2 - Beta(1, 1)						1064.7
U3 - Beta(0.3, 0.7)						1026.7
U3 - Beta(0.5, 0.5)						1058.7
U3 - Beta(0.7, 0.3)						1064.7
U3 - Beta(1, 1)						1058.6

Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.5	34.4	2.5	0	0	50.6
U2 - Beta(0.5, 0.5)	15.1	31.1	3.4	0.1	0	50.3
U2 - Beta(0.7, 0.3)	15.5	32.4	3	0.1	0	49
U2 - Beta(1, 1)	14.8	32.3	3.1	0.1	0	49.7
U3 - Beta(0.3, 0.7)	12.4	33.5	2.5	0	0	51.6
U3 - Beta(0.5, 0.5)	15.2	32.3	2.6	0	0	49.9
U3 - Beta(0.7, 0.3)	16.8	32.3	2.8	0.2	0	47.9
U3 - Beta(1, 1)	16.1	32.6	2.1	0.1	0	49.1
Mean administrations						
U2 - Beta(0.3, 0.7)	10.5	11.2	1.6	0.1	0	-
U2 - Beta(0.5, 0.5)	10.6	10	2.6	0.6	0	-
U2 - Beta(0.7, 0.3)	10.6	9.7	2.8	0.9	0	-
U2 - Beta(1, 1)	10.5	10	2.7	0.6	0	-
U3 - Beta(0.3, 0.7)	10.6	11.2	1.5	0.1	0	-
U3 - Beta(0.5, 0.5)	10.7	10	2.5	0.6	0	-
U3 - Beta(0.7, 0.3)	10.7	9.8	2.7	0.7	0	-
U3 - Beta(1, 1)	10.7	10	2.5	0.6	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.7	4.9	0.6	0.1	0	-
U2 - Beta(0.5, 0.5)	2.7	4.5	0.8	0.3	0	-
U2 - Beta(0.7, 0.3)	2.7	4.4	0.8	0.3	0	-
U2 - Beta(1, 1)	2.6	4.5	0.8	0.3	0	-
U3 - Beta(0.3, 0.7)	2.8	4.9	0.5	0.1	0	-
U3 - Beta(0.5, 0.5)	2.7	4.5	0.8	0.2	0	-
U3 - Beta(0.7, 0.3)	2.7	4.4	0.8	0.3	0	-
U3 - Beta(1, 1)	2.7	4.5	0.8	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						835.2
U2 - Beta(0.5, 0.5)						853.4
U2 - Beta(0.7, 0.3)						860.8
U2 - Beta(1, 1)						856
U3 - Beta(0.3, 0.7)						828.5
U3 - Beta(0.5, 0.5)						852
U3 - Beta(0.7, 0.3)						859.6
U3 - Beta(1, 1)						853.5

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	84.8	12.1	0.4	0	0	2.7
U2 - Beta(0.5, 0.5)	67.7	27.9	1.5	0	0	2.9
U2 - Beta(0.7, 0.3)	67	28.4	2	0	0	2.6
U2 - Beta(1, 1)	68.4	26.9	2	0	0	2.7
U3 - Beta(0.3, 0.7)	85.3	12	0	0	0	2.7
U3 - Beta(0.5, 0.5)	72	24.4	0.8	0	0	2.8
U3 - Beta(0.7, 0.3)	69.3	25.3	2.8	0	0	2.6
U3 - Beta(1, 1)	70.8	25.3	1	0.1	0	2.8
Mean administrations						
U2 - Beta(0.3, 0.7)	25.7	3.5	0.2	0	0	-
U2 - Beta(0.5, 0.5)	19	8.7	1.5	0.2	0	-
U2 - Beta(0.7, 0.3)	18.1	8.8	2	0.5	0	-
U2 - Beta(1, 1)	19.1	8.7	1.5	0.2	0	-
U3 - Beta(0.3, 0.7)	25.9	3.4	0.1	0	0	-
U3 - Beta(0.5, 0.5)	20.1	8.2	1.1	0.1	0	-
U3 - Beta(0.7, 0.3)	18.3	9	1.8	0.4	0	-
U3 - Beta(1, 1)	20	8.2	1	0.1	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	7.3	1.4	0.1	0	0	-
U2 - Beta(0.5, 0.5)	4.7	3.2	0.5	0.1	0	-
U2 - Beta(0.7, 0.3)	4.5	3.1	0.6	0.1	0	-
U2 - Beta(1, 1)	4.7	3.2	0.5	0.1	0	-
U3 - Beta(0.3, 0.7)	7.4	1.3	0	0	0	-
U3 - Beta(0.5, 0.5)	5.1	3.1	0.3	0	0	-
U3 - Beta(0.7, 0.3)	4.5	3.2	0.6	0.1	0	-
U3 - Beta(1, 1)	5.1	3.1	0.3	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1144.1
U2 - Beta(0.5, 0.5)						1158.7
U2 - Beta(0.7, 0.3)						1169.6
U2 - Beta(1, 1)						1159.5
U3 - Beta(0.3, 0.7)						1143.8
U3 - Beta(0.5, 0.5)						1156.9
U3 - Beta(0.7, 0.3)						1164.9
U3 - Beta(1, 1)						1156
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.2	15.1	23.3	48.4	3.6	0.4
U2 - Beta(0.5, 0.5)	15.9	12.3	17	46.3	8	0.5
U2 - Beta(0.7, 0.3)	19.7	16.6	15.7	40	7.6	0.4
U2 - Beta(1, 1)	17.4	13.3	15	45.9	8	0.4
U3 - Beta(0.3, 0.7)	15.6	19.9	22.2	40.2	1.6	0.5
U3 - Beta(0.5, 0.5)	21	15.7	15.5	40.2	7.1	0.5
U3 - Beta(0.7, 0.3)	23.2	18.4	13.9	37.4	6.7	0.4
U3 - Beta(1, 1)	23.5	18.5	13.2	38.6	5.8	0.4
Mean administrations						
U2 - Beta(0.3, 0.7)	6	7.4	7.9	7.9	0.8	-
U2 - Beta(0.5, 0.5)	4.6	5.5	6.5	10.2	3.1	-
U2 - Beta(0.7, 0.3)	4.8	5.5	6.6	9.3	3.8	-
U2 - Beta(1, 1)	4.7	5.6	6.5	10.1	3.1	-
U3 - Beta(0.3, 0.7)	6.8	8.3	7.6	6.8	0.5	-
U3 - Beta(0.5, 0.5)	4.9	5.8	6.5	10	2.7	-
U3 - Beta(0.7, 0.3)	5.2	5.9	6.5	9.2	3.2	-
U3 - Beta(1, 1)	5	5.9	6.4	9.9	2.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.4	3.8	3	2.7	0.3	-
U2 - Beta(0.5, 0.5)	0.7	3.2	2.1	3.4	1	-
U2 - Beta(0.7, 0.3)	0.8	3.2	2	3.1	1.1	-
U2 - Beta(1, 1)	0.7	3.2	2.1	3.4	1	-
U3 - Beta(0.3, 0.7)	1.8	4.1	3	2.4	0.2	-
U3 - Beta(0.5, 0.5)	0.8	3.3	2.1	3.4	0.9	-
U3 - Beta(0.7, 0.3)	1	3.4	2	3.1	1	-
U3 - Beta(1, 1)	0.9	3.4	2	3.3	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1029.9
U2 - Beta(0.5, 0.5)						1076
U2 - Beta(0.7, 0.3)						1080.5
U2 - Beta(1, 1)						1074.6
U3 - Beta(0.3, 0.7)						1018.2
U3 - Beta(0.5, 0.5)						1069.6
U3 - Beta(0.7, 0.3)						1068.9
U3 - Beta(1, 1)						1069

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.4	77.6	11.5	1.5	0	0
U2 - Beta(0.5, 0.5)	1.7	45	32.8	18.5	2	0
U2 - Beta(0.7, 0.3)	5.1	36.5	25.4	27	6	0
U2 - Beta(1, 1)	1.6	45.5	32.7	18.5	1.7	0
U3 - Beta(0.3, 0.7)	12.7	78	8.4	0.9	0	0
U3 - Beta(0.5, 0.5)	3.8	50.2	28.7	16.4	0.9	0
U3 - Beta(0.7, 0.3)	4.6	40.8	26.8	24.4	3.4	0
U3 - Beta(1, 1)	3.5	50.3	29.6	15.7	0.9	0
Mean administrations						
U2 - Beta(0.3, 0.7)	9	18.5	2.2	0.2	0	-
U2 - Beta(0.5, 0.5)	3.9	14.3	7.6	3.6	0.5	-
U2 - Beta(0.7, 0.3)	3.7	7.4	7.8	8	3.1	-
U2 - Beta(1, 1)	3.9	14.4	7.6	3.6	0.5	-
U3 - Beta(0.3, 0.7)	9.4	18.7	1.7	0.2	0	-
U3 - Beta(0.5, 0.5)	4.1	15.2	7.1	3.2	0.4	-
U3 - Beta(0.7, 0.3)	3.9	9.7	7.9	6.6	1.9	-
U3 - Beta(1, 1)	4.1	15.2	7.2	3.2	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.6	6.7	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	0.3	5.4	2.4	1.1	0.2	-
U2 - Beta(0.7, 0.3)	0.2	3.3	2.2	2.2	0.9	-
U2 - Beta(1, 1)	0.3	5.5	2.3	1.1	0.2	-
U3 - Beta(0.3, 0.7)	2.8	6.7	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	0.4	5.8	2.2	1	0.1	-
U3 - Beta(0.7, 0.3)	0.3	4	2.3	1.9	0.6	-
U3 - Beta(1, 1)	0.4	5.8	2.2	1	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1097.1
U2 - Beta(0.5, 0.5)						1137.8
U2 - Beta(0.7, 0.3)						1164.9
U2 - Beta(1, 1)						1139
U3 - Beta(0.3, 0.7)						1094.8
U3 - Beta(0.5, 0.5)						1135.8
U3 - Beta(0.7, 0.3)						1152.6
U3 - Beta(1, 1)						1134.8
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.7	9.2	50.2	31.2	5.6	0.1
U2 - Beta(0.5, 0.5)	10.8	14	35.2	31.5	8.3	0.2
U2 - Beta(0.7, 0.3)	12.9	17.8	32.8	27.9	8.5	0.1
U2 - Beta(1, 1)	12.2	16.3	33.7	29.3	8.4	0.1
U3 - Beta(0.3, 0.7)	7	12.4	49.7	27.7	3.1	0.1
U3 - Beta(0.5, 0.5)	15.3	16.2	35	27.4	6	0.1
U3 - Beta(0.7, 0.3)	16.4	21	33.3	23.8	5.4	0.1
U3 - Beta(1, 1)	18	19.1	32.6	25.3	4.9	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.3	6.6	12.1	5.8	1.2	-
U2 - Beta(0.5, 0.5)	3.5	4.6	8.9	8.9	4.1	-
U2 - Beta(0.7, 0.3)	3.6	4.4	8.6	8.7	4.7	-
U2 - Beta(1, 1)	3.5	4.6	9	8.9	4.1	-
U3 - Beta(0.3, 0.7)	4.6	7.1	12.1	5.4	0.8	-
U3 - Beta(0.5, 0.5)	3.7	4.8	9.3	8.6	3.5	-
U3 - Beta(0.7, 0.3)	3.8	5	8.9	8.4	3.9	-
U3 - Beta(1, 1)	3.8	5.1	9.2	8.5	3.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.6	4.6	2.1	0.4	-
U2 - Beta(0.5, 0.5)	0.2	3	2.9	3.3	1.4	-
U2 - Beta(0.7, 0.3)	0.3	2.9	2.8	3.3	1.5	-
U2 - Beta(1, 1)	0.2	2.9	2.9	3.4	1.3	-
U3 - Beta(0.3, 0.7)	0.8	3.8	4.6	2	0.3	-
U3 - Beta(0.5, 0.5)	0.3	3.1	3.1	3.3	1.2	-
U3 - Beta(0.7, 0.3)	0.3	3.2	2.9	3.2	1.3	-
U3 - Beta(1, 1)	0.4	3.2	3	3.2	1.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1022.1
U2 - Beta(0.5, 0.5)						1052.7
U2 - Beta(0.7, 0.3)						1058.7
U2 - Beta(1, 1)						1053
U3 - Beta(0.3, 0.7)						1018.6
U3 - Beta(0.5, 0.5)						1048.7
U3 - Beta(0.7, 0.3)						1052.3
U3 - Beta(1, 1)						1046.7

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.7	3.9	20.8	61	10.5	0.1
U2 - Beta(0.5, 0.5)	8.8	5.7	13.2	51.6	20.6	0.1
U2 - Beta(0.7, 0.3)	9.4	7.4	13.1	46	24	0.1
U2 - Beta(1, 1)	8.6	6.1	12.4	51.4	21.4	0.1
U3 - Beta(0.3, 0.7)	6.5	4.2	23	60.1	6.1	0.1
U3 - Beta(0.5, 0.5)	12.6	7.4	12.1	50.3	17.5	0.1
U3 - Beta(0.7, 0.3)	12.9	9.3	12.2	46.4	19.1	0.1
U3 - Beta(1, 1)	13.3	9.2	11.5	49.4	16.5	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.8	4.5	8.6	11.3	1.8	-
U2 - Beta(0.5, 0.5)	3.3	3.8	6.3	11.2	5.3	-
U2 - Beta(0.7, 0.3)	3.4	3.8	6.2	10.1	6.6	-
U2 - Beta(1, 1)	3.4	3.9	6.2	11.2	5.4	-
U3 - Beta(0.3, 0.7)	3.8	4.7	9.1	11.2	1.3	-
U3 - Beta(0.5, 0.5)	3.5	3.9	6.4	11.4	4.8	-
U3 - Beta(0.7, 0.3)	3.6	4	6.2	10.4	5.8	-
U3 - Beta(1, 1)	3.6	4.2	6.2	11.2	4.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.8	3.3	3.9	0.6	-
U2 - Beta(0.5, 0.5)	0.2	2.7	1.9	4	1.5	-
U2 - Beta(0.7, 0.3)	0.2	2.8	1.8	3.7	1.7	-
U2 - Beta(1, 1)	0.2	2.8	1.8	3.9	1.5	-
U3 - Beta(0.3, 0.7)	0.4	2.9	3.4	3.9	0.4	-
U3 - Beta(0.5, 0.5)	0.2	2.8	1.9	4.1	1.3	-
U3 - Beta(0.7, 0.3)	0.3	2.9	1.8	3.7	1.6	-
U3 - Beta(1, 1)	0.3	2.9	1.9	4	1.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1048.4
U2 - Beta(0.5, 0.5)						1087.7
U2 - Beta(0.7, 0.3)						1091.4
U2 - Beta(1, 1)						1087.4
U3 - Beta(0.3, 0.7)						1043.3
U3 - Beta(0.5, 0.5)						1083
U3 - Beta(0.7, 0.3)						1084.2
U3 - Beta(1, 1)						1080.4
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.7	16	56.1	13.6	1.5	5.1
U2 - Beta(0.5, 0.5)	12.8	16.6	48.8	14.9	2.2	4.7
U2 - Beta(0.7, 0.3)	13.1	16.6	47.9	14.5	3.1	4.8
U2 - Beta(1, 1)	12.8	16.3	48.2	15.9	2.4	4.4
U3 - Beta(0.3, 0.7)	10.4	16.4	53.1	13.7	1.1	5.3
U3 - Beta(0.5, 0.5)	16.5	17.9	44.7	14.2	1.8	4.9
U3 - Beta(0.7, 0.3)	17.1	19.4	42.1	15	1.7	4.7
U3 - Beta(1, 1)	18.7	16.5	42.6	15.3	1.7	5.2
Mean administrations						
U2 - Beta(0.3, 0.7)	5.5	7.5	12.5	3.7	0.4	-
U2 - Beta(0.5, 0.5)	5.4	7	10.2	5.4	1.5	-
U2 - Beta(0.7, 0.3)	5.3	7	9.7	5.7	1.9	-
U2 - Beta(1, 1)	5.4	7	10.2	5.4	1.5	-
U3 - Beta(0.3, 0.7)	5.9	7.8	12.1	3.5	0.3	-
U3 - Beta(0.5, 0.5)	5.7	7.2	10.2	5.2	1.3	-
U3 - Beta(0.7, 0.3)	5.7	7.2	9.5	5.5	1.6	-
U3 - Beta(1, 1)	6	7.3	9.9	5.2	1.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.1	3.8	4.3	1.4	0.1	-
U2 - Beta(0.5, 0.5)	1	3.8	3.2	2	0.5	-
U2 - Beta(0.7, 0.3)	1	3.9	3	2	0.6	-
U2 - Beta(1, 1)	1.1	3.8	3.2	2	0.5	-
U3 - Beta(0.3, 0.7)	1.3	3.9	4.2	1.3	0.1	-
U3 - Beta(0.5, 0.5)	1.2	3.9	3.2	1.9	0.4	-
U3 - Beta(0.7, 0.3)	1.2	4	3	1.9	0.6	-
U3 - Beta(1, 1)	1.3	4	3.1	1.9	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1029.9
U2 - Beta(0.5, 0.5)						1044.9
U2 - Beta(0.7, 0.3)						1046.9
U2 - Beta(1, 1)						1043.9
U3 - Beta(0.3, 0.7)						1027.3
U3 - Beta(0.5, 0.5)						1042.2
U3 - Beta(0.7, 0.3)						1040
U3 - Beta(1, 1)						1037.9

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.5	74.1	14.3	1	0	0.1
U2 - Beta(0.5, 0.5)	6.7	48.2	34.9	9	1.1	0.1
U2 - Beta(0.7, 0.3)	7.7	45.3	32.8	13	1.1	0.1
U2 - Beta(1, 1)	6.5	48.6	35.2	8.4	1.2	0.1
U3 - Beta(0.3, 0.7)	14	74.7	10.4	0.8	0	0.1
U3 - Beta(0.5, 0.5)	8.3	53.7	30.7	6.5	0.7	0.1
U3 - Beta(0.7, 0.3)	10.7	49.5	28.2	10.4	1.1	0.1
U3 - Beta(1, 1)	9.1	56.1	28.1	6.1	0.5	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	9.2	17.8	2.8	0.2	0	-
U2 - Beta(0.5, 0.5)	4.4	13.8	8.4	2.8	0.5	-
U2 - Beta(0.7, 0.3)	4.4	10.1	8.9	5.1	1.5	-
U2 - Beta(1, 1)	4.4	13.8	8.5	2.7	0.5	-
U3 - Beta(0.3, 0.7)	9.6	18	2.2	0.2	0	-
U3 - Beta(0.5, 0.5)	4.7	14.7	7.9	2.3	0.4	-
U3 - Beta(0.7, 0.3)	4.7	11.8	8.7	4	0.8	-
U3 - Beta(1, 1)	4.7	15	7.7	2.2	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.7	6.6	0.9	0.1	0	-
U2 - Beta(0.5, 0.5)	0.5	5.3	2.4	0.9	0.2	-
U2 - Beta(0.7, 0.3)	0.5	4.1	2.4	1.5	0.4	-
U2 - Beta(1, 1)	0.5	5.3	2.4	0.9	0.2	-
U3 - Beta(0.3, 0.7)	2.9	6.7	0.7	0.1	0	-
U3 - Beta(0.5, 0.5)	0.6	5.7	2.2	0.7	0.1	-
U3 - Beta(0.7, 0.3)	0.6	4.7	2.4	1.2	0.2	-
U3 - Beta(1, 1)	0.6	5.8	2.1	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1086.8
U2 - Beta(0.5, 0.5)						1144.3
U2 - Beta(0.7, 0.3)						1160.9
U2 - Beta(1, 1)						1143.4
U3 - Beta(0.3, 0.7)						1083.8
U3 - Beta(0.5, 0.5)						1137.8
U3 - Beta(0.7, 0.3)						1149.2
U3 - Beta(1, 1)						1135.9
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.4	39.8	13.1	2.6	0.6	30.5
U2 - Beta(0.5, 0.5)	15.8	40.4	11.5	3.6	0.5	28.2
U2 - Beta(0.7, 0.3)	15.5	40.3	12.9	3	0.6	27.7
U2 - Beta(1, 1)	17.2	38.6	12.3	2.9	0.5	28.5
U3 - Beta(0.3, 0.7)	15.7	40.8	10.3	2.2	0.3	30.7
U3 - Beta(0.5, 0.5)	17.4	40.3	10.2	2.7	0.3	29.1
U3 - Beta(0.7, 0.3)	17.3	40.7	10.3	2.6	0.3	28.8
U3 - Beta(1, 1)	19.3	38.7	9.9	3.2	0.3	28.6
Mean administrations						
U2 - Beta(0.3, 0.7)	9	11.5	4.4	1.4	0.2	-
U2 - Beta(0.5, 0.5)	8.8	10.3	4.8	2.4	0.5	-
U2 - Beta(0.7, 0.3)	8.8	10	4.8	2.6	0.6	-
U2 - Beta(1, 1)	8.8	10.3	4.8	2.4	0.5	-
U3 - Beta(0.3, 0.7)	9.4	11.7	4	1.2	0.1	-
U3 - Beta(0.5, 0.5)	9.1	10.3	4.5	2.3	0.5	-
U3 - Beta(0.7, 0.3)	9	10.3	4.6	2.4	0.5	-
U3 - Beta(1, 1)	9.1	10.5	4.4	2.3	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.2	5.4	1.6	0.6	0.1	-
U2 - Beta(0.5, 0.5)	2.1	5	1.5	0.9	0.2	-
U2 - Beta(0.7, 0.3)	2.1	4.9	1.5	1	0.2	-
U2 - Beta(1, 1)	2.1	5	1.5	0.9	0.2	-
U3 - Beta(0.3, 0.7)	2.4	5.5	1.4	0.5	0.1	-
U3 - Beta(0.5, 0.5)	2.2	5	1.4	0.9	0.2	-
U3 - Beta(0.7, 0.3)	2.2	5.1	1.4	0.9	0.2	-
U3 - Beta(1, 1)	2.2	5.2	1.4	0.9	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						916
U2 - Beta(0.5, 0.5)						938
U2 - Beta(0.7, 0.3)						938.8
U2 - Beta(1, 1)						937
U3 - Beta(0.3, 0.7)						909.1
U3 - Beta(0.5, 0.5)						930.8
U3 - Beta(0.7, 0.3)						937.1
U3 - Beta(1, 1)						932.3

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - dose-specific alpha efficacy - IPDE alpha = 0.6 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	15.4	47	12.9	1.4	0.1	23.2
U2 - Beta(0.5, 0.5)	15.2	48.9	11.1	1	0	23.8
U2 - Beta(0.7, 0.3)	15.3	48.6	11.7	1.8	0	22.6
U2 - Beta(1, 1)	15.7	48	11.7	1.6	0	23
U3 - Beta(0.3, 0.7)	17.3	48.4	9.4	0.9	0.1	23.9
U3 - Beta(0.5, 0.5)	17.1	48.2	10.8	1	0	22.9
U3 - Beta(0.7, 0.3)	16.7	48.4	10.9	1.6	0	22.4
U3 - Beta(1, 1)	18.8	46.3	10.1	1.6	0	23.2
Mean administrations						
U2 - Beta(0.3, 0.7)	8.6	13	4.6	1.2	0.1	-
U2 - Beta(0.5, 0.5)	8.6	11.7	4.9	2.1	0.4	-
U2 - Beta(0.7, 0.3)	8.5	11.5	5	2.3	0.5	-
U2 - Beta(1, 1)	8.5	11.8	4.9	2.1	0.4	-
U3 - Beta(0.3, 0.7)	9	13.3	4.1	1	0.1	-
U3 - Beta(0.5, 0.5)	8.9	11.7	4.7	2	0.4	-
U3 - Beta(0.7, 0.3)	8.7	11.7	4.7	2.1	0.4	-
U3 - Beta(1, 1)	8.9	11.7	4.6	2	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	6	1.7	0.5	0.1	-
U2 - Beta(0.5, 0.5)	2	5.6	1.5	0.9	0.1	-
U2 - Beta(0.7, 0.3)	2	5.6	1.5	0.9	0.2	-
U2 - Beta(1, 1)	2	5.7	1.5	0.9	0.2	-
U3 - Beta(0.3, 0.7)	2.3	6	1.5	0.4	0	-
U3 - Beta(0.5, 0.5)	2.2	5.7	1.4	0.8	0.1	-
U3 - Beta(0.7, 0.3)	2.1	5.7	1.4	0.9	0.1	-
U3 - Beta(1, 1)	2.2	5.6	1.4	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						948.2
U2 - Beta(0.5, 0.5)						959.2
U2 - Beta(0.7, 0.3)						963.4
U2 - Beta(1, 1)						962.4
U3 - Beta(0.3, 0.7)						944.4
U3 - Beta(0.5, 0.5)						958.3
U3 - Beta(0.7, 0.3)						959.3
U3 - Beta(1, 1)						958

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.